

## ESSENTIAL MATHEMATICS GRADE 10: AREA OF PART OF A CIRCLE

*CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.5 (8 Lessons)*

| SUB-STRAND OVERVIEW             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| Grade Level                     | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Subject                         | Essential Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Strand                          | Strand 2.0: Measurements and Geometry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Sub-Strand                      | Sub-Strand 2.5: Area of Part of a Circle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Total Duration                  | 8 lessons × 40 minutes = 320 minutes total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Sub-Strand Content              | <ul style="list-style-type: none"> <li>– Area of a sector — deriving and applying the formula (<math>\theta/360 \times \pi r^2</math>) using paper cut-outs of sectors with measured radius and included angle</li> <li>– Area of a segment — computing area of a segment by subtracting the area of the triangle from the area of the corresponding sector</li> <li>– Applications of area of part of a circle in real-life situations — designing patterns, cutting circular fruits (e.g. watermelon), area swept by an opening door, decorative surfaces</li> </ul>                                                                                                                                                                                                                                                  |
| Learning Outcomes               | <p>By the end of the sub-strand, the learner should be able to:</p> <ol style="list-style-type: none"> <li>determine the area of a sector of a circle in different situations,</li> <li>determine the area of a segment of a circle in different situations,</li> <li>apply the area of part of a circle in real-life situations,</li> <li>appreciate the application of area of part of circular objects in real-life situations.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                           |
| Core Competencies               | <ul style="list-style-type: none"> <li>– Critical Thinking and Problem Solving: the learner explores different options of improvising cut-outs of sectors of a circle for working out their areas, and selects appropriate formulas when solving problems involving sectors and segments.</li> <li>– Learning to Learn: the learner acquires new knowledge as they discuss with peers how to work out areas of circular objects, and independently investigates the relationship between a sector and its corresponding triangle to derive the segment area formula.</li> <li>– Citizenship: learners appreciate their immediate environment as they gather locally available circular objects (e.g. sufuria lids, clock faces, wheel hubs) and use them to explore real-life applications of circular area.</li> </ul> |
| Core Values                     | <ul style="list-style-type: none"> <li>– Unity: The learner exhibits devotion and honesty by undertaking tasks while working with peers to make paper cut-outs of sectors of a circle and obtain measurements, sharing resources equitably within the group.</li> <li>– Peace: the learner displays care and tolerance while discussing with peers and obtaining circular objects, tracing and marking out sectors and segments, and respecting differing approaches to problem-solving.</li> </ul>                                                                                                                                                                                                                                                                                                                     |
| Science & Engineering Practices | <ul style="list-style-type: none"> <li>– Asking Questions and Defining Problems: learners pose questions about why a slice of watermelon or a swept floor area is not a simple rectangle or triangle, driving investigation into circular area formulas.</li> <li>– Planning and Carrying Out Investigations: learners design and execute a hands-on cut-out activity — measuring radius and central angle of paper sectors, recording data in a table, and computing areas to verify the formula.</li> <li>– Using Mathematics and Computational Thinking: learners apply the sector and segment area formulas (<math>\theta/360 \times \pi r^2</math>; sector</li> </ul>                                                                                                                                              |

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|                                                   | <p>area minus triangle area) to solve quantitative problems set in Kenyan contexts.</p> <ul style="list-style-type: none"> <li>– Constructing Explanations: learners articulate why the formula for a sector is a fraction of the full circle's area, linking the central angle to the proportion of the circle, and explain the segment derivation step-by-step.</li> <li>– Obtaining, Evaluating, and Communicating Information: learners present their findings on real-life applications (door sweeps, watermelon slices, wiper blade arcs) to peers and evaluate each other's reasoning.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Pertinent &amp; Contemporary Issues (PCIs)</b> | <ul style="list-style-type: none"> <li>– Education for Sustainable Development: the learner uses locally available circular objects (sufuria lids, tin can bases, clock faces found in Nairobi markets) to trace and mark out sectors and segments, minimising waste by repurposing materials as learning resources.</li> <li>– Life Skills (Decision Making): the learner makes informed decisions when selecting appropriate formulas and methods for calculating the area of a sector or segment in varied real-life problem contexts.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Career Connections</b>                         | <ul style="list-style-type: none"> <li>– Architecture and Interior Design: calculating the floor area swept by doors and designing circular decorative patterns on buildings in cities like Nairobi and Mombasa requires accurate sector and segment area computations.</li> <li>– Hospitality and Catering: portioning circular foods (watermelon, mandazi rounds, cake) into equal or specified sector-shaped servings requires understanding sector angles and areas — directly relevant to Kenya's hospitality industry.</li> <li>– Sports Management and Coaching: computing the area of circular sectors on athletic tracks and field event circles (e.g. discus throwing sector) at venues such as Nyayo Stadium uses sector area concepts.</li> <li>– Journalism and Photography: framing circular graphic elements, pie-chart overlays, and camera aperture calculations in media production connect to arc and sector proportions.</li> <li>– Agriculture and Land Management: parcelling circular irrigation plots or rotational grazing paddocks in the Rift Valley into sector-shaped portions requires applying area of a sector to real land areas.</li> </ul> |
| <b>Focus for Lessons</b>                          | <p>This unit develops learners' ability to calculate and apply the area of sectors and segments of a circle, building directly on prior knowledge of circle properties and polygon areas (Sub-Strand 2.4). Learners progress from hands-on paper cut-out investigations to abstract formula derivation and finally to solving authentic Kenyan real-life problems. The unit foregrounds collaborative inquiry, local material use, and the utility of mathematics in everyday contexts such as food portioning, door clearance planning, and decorative design.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Driving Question / Key Inquiry Questions</b>   | <p>DRIVING QUESTION: How can we calculate the exact area of any wedge-shaped or segment-shaped piece of a circle — like a watermelon slice at Wakulima Market — so that we can solve fair-sharing and design problems in real life?</p> <p>KICD KEY INQUIRY QUESTION:</p> <p>1. Where do we use area of part of a circle in real life?</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Anchoring Phenomenon</b>                       | <p>A mama mboga at Wakulima Market in Nairobi slices a large, perfectly circular watermelon into unequal wedge-shaped pieces to sell by the slice. Some customers complain that their pieces look smaller than others even though the vendor claims she cut the melon fairly. A student notices that two pieces cut at different angles but with the same radius look very different in size — one is a thin sliver and another is a wide wedge. The puzzle: the pieces are all bounded by two straight cuts and a curved rind, yet their sizes are clearly different. Why does changing only the angle of the cut change the area of the piece so dramatically? How can we calculate exactly how much watermelon (in <math>\text{cm}^2</math>) is in any given wedge so that every customer truly gets a fair share? This is puzzling because learners' prior experience with area (rectangles, triangles, polygons) gives them no direct method for handling a shape bounded by two radii and a curved arc.</p>                                                                                                                                                             |
| <b>Storyline Thread</b>                           | <p>Lesson 1 — PREDICT: Anchoring Phenomenon Launch (Sector intuition)</p> <p>Learners are shown a large paper circle representing a watermelon cross-section. The teacher poses the driving question and asks learners to predict (individually, in writing) which of three drawn sectors — with angles <math>45^\circ</math>, <math>90^\circ</math>, and <math>135^\circ</math> and the same radius —</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

has the largest area and by approximately how much. Cold-call 4 learners to share predictions; record on the Driving Question Board (DQB). Learners then discuss in pairs: what information would you need to calculate the exact area of each slice? Introduce vocabulary: sector, radius, central angle, arc.

#### Lesson 2 — OBSERVE: Investigating Sector Area with Cut-Outs

Learners work in groups of 4. Each group receives a printed circle (radius 7 cm), a protractor, scissors, and a ruler. Groups cut out sectors of specified angles (30°, 60°, 90°, 120°, 180°) and arrange them side by side. They measure and record radius and angle, then calculate each sector's area as a fraction of the full circle area ( $\pi r^2$ ). Groups tabulate results and identify the pattern: Area of sector =  $(\theta/360) \times \pi r^2$ . Teacher circulates, asking targeted questions: "What fraction of the full circle is your 90° sector?" WAIT TIME: 30 seconds after each question before cold-calling.

#### Lesson 3 — EXPLAIN: Deriving and Applying the Sector Area Formula

Teacher formalises the sector area formula from the groups' findings. Learners work through three graduated examples in Kenyan contexts: (1) a 120° slice of a watermelon of radius 15 cm; (2) the area swept by a school gate of radius 1.2 m swinging through 75°; (3) a centre-pivot irrigation arc of radius 80 m through 270°. Pairs solve problems and present solutions; class evaluates. Learners update the DQB with the sector formula and one new question that has arisen.

#### Lesson 4 — OBSERVE: Investigating Segment Area

Learners return to their sector cut-outs. Teacher guides: "Inside this sector, there is a triangle formed by the two radii and the chord. What shape is left if we remove that triangle?" Learners cut out the triangle (base = chord, two equal sides = radius) from the sector to isolate the segment. They measure dimensions, calculate triangle area ( $\frac{1}{2}r^2\sin\theta$ ), and subtract from sector area to find segment area. Groups record the formula: Area of segment =  $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ . Observation schedule used for group work assessment.

#### Lesson 5 — EXPLAIN: Applying the Segment Area Formula

Teacher works through two fully worked examples: (1) finding the area of the segment of a circle of radius 10 cm with a chord subtending 60° at the centre (context: leather coaster waste in Kericho); (2) the area of glass NOT cleaned by a matatu wiper of radius 30 cm that sweeps 110° (the uncleaned segment near the pivot). Learners then attempt two problems independently. WAIT TIME: allow 5 minutes silent work before peer discussion. Cold-call 3 learners to present solutions on the board. Correct misconceptions around the subtraction step.

#### Lesson 6 — DRIVING QUESTION BOARD (DQB) Update + Mixed Practice

Learners revisit the DQB, answer questions resolved so far (sector formula, segment formula), and post new "wonder" questions. Teacher facilitates a class discussion linking back to the anchoring phenomenon: "Can we now calculate whether the mama mboga's slices are truly equal?" Learners calculate and compare three watermelon slice areas (angles 40°, 55°, 75°; radius 18 cm) and write a short justification of fairness. Introduce mixed problems combining both sector and segment in one context (e.g. total area of a decorative window panel in a Nairobi mosque).

#### Lesson 7 — MODEL BUILDING: Real-Life Design Task

Learners work in pairs on a design challenge: plan a circular flowerbed of radius 2 m in the school compound that is divided into 4 sector-shaped sections for different plants; one section must be exactly 25% of the total area, two sections must be equal, and the fourth section is the remainder. Learners must calculate and label the angle and area of each section. Extension: one pair calculates the area of a decorative stone path that forms a segment around the outer edge of one section. Learners produce a

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|                      | <p>labelled diagram with all calculations shown.</p> <p>Lesson 8 — SUMMATIVE REVIEW + MODEL PRESENTATION</p> <p>Groups present their flowerbed designs and calculations to the class; peers provide structured feedback using the KICD assessment rubric criteria (sector area, segment area, application). Teacher conducts a whole-class consolidation: revisit the driving question with final answers, review the DQB for any unresolved questions, and administer a short written class activity (4 questions: 1 sector, 1 segment, 1 application, 1 multi-step). Learners reflect in their portfolios on what changed in their thinking from Lesson 1 to Lesson 8 regarding the watermelon phenomenon.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Supporting Phenomena | <ul style="list-style-type: none"> <li>– A watchman at a school gate in Kisumu notices that the open gate sweeps a fan-shaped region on the ground as it swings from closed to fully open — the painted warning zone on the ground is sector-shaped; how much ground area must be kept clear?</li> <li>– A matatu windscreen wiper in Mombasa clears a sector-shaped region of glass with each sweep; the driver wants to know how much of the windscreen is actually cleaned and how much remains obscured — this involves the area of a sector (and of the uncleaned segment near the pivot).</li> <li>– A craftsperson in Kericho making decorative leather coasters cuts circular discs and then removes a circular segment from each to create a "D-shaped" coaster — calculating how much leather is wasted per coaster requires segment area.</li> <li>– A farmer in the Rift Valley installs a centre-pivot irrigation system that rotates through a <math>270^\circ</math> arc; the county water office needs to know the irrigated area in hectares to allocate water rights — this is a direct sector-area calculation on a large scale.</li> </ul> |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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## LESSON 1 (40 minutes): Anchoring Phenomenon Launch — Sector Intuition at Wakulima Market

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Purpose</b>              | To launch the anchoring phenomenon by confronting learners with the puzzle of why watermelon wedges cut from the same circular melon at different angles have dramatically different areas, and to surface prior knowledge and initial intuitions about area.                                                                                                                                |
| <b>Knowledge</b>            | <ul style="list-style-type: none"> <li>– A sector is the region bounded by two radii and an arc of a circle.</li> <li>– The central angle (angle at the centre) is the key variable that determines the size of a sector when the radius is fixed.</li> <li>– Prior area methods (rectangles, triangles, polygons) cannot be directly applied to shapes bounded by a curved edge.</li> </ul> |
| <b>Skills</b>               | <ul style="list-style-type: none"> <li>– Estimate and compare areas of sector-shaped pieces using reasoning and visual inspection.</li> <li>– Formulate mathematical questions arising from a real-life fair-sharing problem.</li> </ul>                                                                                                                                                     |
| <b>Attitudes</b>            | <ul style="list-style-type: none"> <li>– Appreciate that mathematics is embedded in everyday Kenyan market activities.</li> <li>– Show curiosity and intellectual honesty when prior knowledge is insufficient to solve a new problem.</li> </ul>                                                                                                                                            |
| <b>Key Inquiry Question</b> | Where do we use area of part of a circle in real life?                                                                                                                                                                                                                                                                                                                                       |
| <b>Purpose in Storyline</b> | This is the opening lesson that anchors the entire sub-strand in the Wakulima Market watermelon phenomenon; it surfaces learners' intuitions about sector area, exposes the gap in their prior knowledge, and launches the Driving Question Board that will be built upon throughout the sub-strand.                                                                                         |
| <b>Safety Notes</b>         | No specific hazards. If a physical watermelon or knife is brought to class for demonstration, the teacher only handles the cutting implement; learners observe from their seats and do not handle any sharp objects.                                                                                                                                                                         |

### B. LESSON OVERVIEW

This lesson opens the Area of Part of a Circle sub-strand by immersing Grade 10 learners in a vivid, locally grounded phenomenon: a mama mboga at Wakulima Market in Nairobi slicing a large circular watermelon into unequal wedge-shaped pieces. Some customers complain that their pieces look smaller, yet the vendor insists the cuts are fair. Learners are immediately invited to take a position — before any instruction — on which piece is larger and why. This predict-first approach deliberately surfaces the misconception that equal-looking cuts are equal in area, and exposes the critical variable of central angle.

In the Observe phase, learners examine printed or projected sector cut-outs drawn on centimetre-square grid paper, all with the same radius (7 cm, a clean number that later connects to the formula  $A = \theta/360 \times \pi r^2$ ) but with central angles of 30°, 90°, 120°, and 180°. By counting grid squares, learners generate raw area estimates for each piece. This concrete counting activity creates a need for a more efficient method and plants the seed that area grows in proportion to the central angle — the key conceptual insight of the sub-strand.

The lesson closes by constructing the class Driving Question Board (DQB), where learners post their burning questions and initial models (rough sketches annotating what they think controls sector size). This DQB will be revisited and updated in every subsequent lesson, giving learners a visible record of their growing understanding. No formula is introduced in Lesson 1; the lesson's sole purpose is to create genuine intellectual need for the mathematics that follows.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>Learners view a projected image (or teacher-drawn sketch on the board) showing a large circular watermelon divided into four wedge-shaped pieces — all with the same radius but with central angles of <math>30^\circ</math>, <math>90^\circ</math>, <math>120^\circ</math>, and <math>180^\circ</math>. Each piece is labelled A, B, C, and D respectively. Individually and silently, learners write in their exercise books: (1) a ranking of the four pieces from smallest to largest area, and (2) one sentence explaining what they think makes one piece bigger than another. After 2 minutes, learners share their ranking with a partner using a Think-Pair-Share protocol before a whole-class discussion.</p> | <p>Display the watermelon diagram and say: 'Mama Wanjiku sells watermelon slices at Wakulima Market every Saturday. She cut this melon into four pieces — A, B, C, and D. A customer is complaining that piece B is much smaller than piece D. Mama Wanjiku says all her cuts are from the same melon, the same size. Who is right? In your book, rank these pieces from smallest to largest and write ONE sentence: what makes a piece big or small?' Allow WAIT TIME of 15 seconds before any learner writes, to let the image settle. After 2 minutes of individual writing, say: 'Turn to your neighbour — 60 seconds — compare your rankings.' After pair discussion, cold-call 4 learners (choose one high-confidence, one hesitant, one who ranked differently from the majority, one who mentioned 'angle' unprompted) using the prompt: 'Tell us your ranking and your reason.' Record all four rankings on the board without evaluating correctness. Then ask the class: 'We have four different rankings on the board. They cannot all be right. What information would help us decide?' Allow 10 seconds WAIT TIME before taking 3 hands.</p> | <p>Think-Pair-Share with public ranking display. By posting all four rankings without judgment, the teacher creates productive controversy — learners can see that the class does not yet have a shared, justified answer. This cognitive dissonance is the engine that drives motivation through the rest of the lesson and the sub-strand.</p>                                                                      | <p>Teacher scans written rankings as learners work (circulate, read 6–8 books). Key indicator: Do learners mention 'angle', 'width of the cut', or 'how open the piece is' in their explanations? Learners who rely only on visual impression ('it looks bigger') have not yet identified the relevant variable. Note these learners for targeted questioning in the Observe phase. Record on a mental checklist: proportion of class who named the central angle as the key variable (expected: fewer than 30% at this stage).</p> |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of a sector</a></p>                                                                                                                                                                                            | <p>Each learner (or pair of learners if resources are limited) receives a printed A4 sheet showing four sector cut-outs drawn accurately on a <math>1\text{ cm}^2</math> grid: Sector A (<math>r = 7\text{ cm}</math>, <math>\theta = 30^\circ</math>), Sector B (<math>r = 7\text{ cm}</math>, <math>\theta = 90^\circ</math>), Sector C (<math>r = 7\text{ cm}</math>, <math>\theta = 120^\circ</math>), and Sector D (<math>r = 7\text{ cm}</math>, <math>\theta = 180^\circ</math>). Learners count grid squares inside each sector (counting a partial square <math>\geq</math> half as 1, and <math>&lt;</math> half as 0) and record their</p>                                                                       | <p>Distribute the grid sheets and say: 'Each of these four shapes represents one watermelon slice. They were all cut from a melon with radius 7 cm — the same melon. Your job right now is NOT to use a formula. Count the squares inside each shape as carefully as you can. Record your estimates in your table.' Circulate for 5 minutes. When most learners have at least two rows complete,</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Structured Data Collection with Guided Noticing. Counting grid squares is a deliberate bridging strategy: it connects the familiar concept of area-as-square-counting to the unfamiliar curved shape, making the area of a sector tangible and concrete. The two guided questions (same/different) direct learners toward the invariant (radius) and the variable (angle), which is the conceptual core of the</p> | <p>Teacher looks for two things during circulation: (1) Are learners' area estimates increasing monotonically from A to D? (Expected: <math>\sim 38\text{ cm}^2</math>, <math>\sim 154\text{ cm}^2</math>, <math>\sim 205\text{ cm}^2</math>, <math>\sim 308\text{ cm}^2</math> for <math>r = 7\text{ cm}</math> — but grid counting will give rough approximations; the key is the ordering.) A learner whose estimates are not increasing has made a counting error or misidentified the sector boundaries</p>                    |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>estimates in a four-row table in their books with columns: Sector   Central Angle (°)   Estimated Area (cm<sup>2</sup>). They then answer two guided questions: 'What stays the same across all four pieces?' and 'What is different?'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>ask a probing question to the whole class (do not wait for all to finish): 'Without telling me the number yet — as you move from Sector A to Sector D, what is happening to the number of squares?' Allow 10 seconds WAIT TIME. Cold-call 3 learners. Then say: 'Finish your table. You have 3 more minutes.' After counting is complete, direct learners to the two guided questions. Cold-call 2 learners for each question. When a learner answers 'the radius is the same', affirm and press: 'Exactly — and what IS different? What did Mama Wanjiku change each time she made a new cut?' Expect and draw out the answer 'the angle' or 'how wide the cut is.' Write on the board: CENTRAL ANGLE — the angle between the two straight cuts.</p>                               | <p>formula <math>A = \theta/360 \times \pi r^2</math>.</p>                                                                                                                                                                                                                                                                                                                                                                               | <p>— intervene immediately. (2) In the guided questions, can learners articulate 'central angle' or an equivalent phrase as the varying quantity? This is the exit criterion for conceptual readiness for Lesson 2.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Area of a circle intuition</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Radians &amp; arc length</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Learners use their four area estimates to fill in a proportional reasoning prompt on the board: 'Sector D has angle 180° and Sector B has angle 90°. Sector D's angle is ___ times bigger than Sector B's angle. Is Sector D's area approximately ___ times bigger than Sector B's area?' Learners check this against their grid counts. They then discuss in groups of three: 'If Mama Wanjiku doubled the angle of a slice — from 60° to 120° — what would happen to the area of the piece? Justify your answer using your table.' Groups share out, and the teacher scribes the emerging conjecture on the board: 'Area of a sector seems to be proportional to the central angle (when the radius is fixed).'</p> | <p>Write the proportional reasoning prompt on the board and say: 'Look at your table. Sector D has angle 180° and Sector B has angle 90°. 180 is exactly how many times 90?' Pause — 5 seconds WAIT TIME — 'Two times. Now look at your area estimates for D and B. Is the area of D approximately two times the area of B? Check it.' Allow 2 minutes. Cold-call 2 learners: 'What did you find?' After confirming the doubling relationship, pose the group discussion question: 'In your group of three — Mama Wanjiku's customer wants DOUBLE the watermelon. She has a 60° slice. What angle should the new slice be, and how does the area change? You have 90 seconds — go.' After group discussion, cold-call one group to present. Scribe their conjecture verbatim, then</p> | <p>Proportional Reasoning Bridge. By asking learners to check whether doubling the angle doubles the area, the teacher is scaffolding the core mathematical structure of the sector area formula (<math>A \propto \theta</math>) without stating it. This is a sense-making move rather than direct instruction — learners arrive at the proportionality conjecture through their own data, which increases ownership and retention.</p> | <p>Teacher listens to group discussions for the quality of justification. Target indicator: Can learners use the word 'proportional' or an equivalent idea ('the area goes up by the same amount as the angle goes up') to justify their conjecture? Learners who say only 'it gets bigger' have not yet made the proportionality connection — the teacher should prompt: 'By how MUCH does it get bigger? Check your numbers.' Also note any group that raises the question of what happens when the radius changes — this shows readiness for extension and should be acknowledged publicly: 'Excellent question — that is something we will explore in a later lesson.'</p> |

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| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of a circle</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Arc length</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>Each learner receives two sticky notes (or tears a page into two pieces if sticky notes are unavailable). On the first note, they write one question they still have about the watermelon problem — something they genuinely do not yet know how to answer. On the second note, they write one thing they are now more sure about after the Observe and Explain phases. Learners post their notes on the class DQB (a dedicated section of the board or a large sheet of paper fixed to the wall), grouping them under two columns: 'Questions We Have' and 'Things We Are Starting to Understand.' The teacher reads several questions aloud and organises them into clusters: questions about the formula, questions about what happens when the radius changes, questions about the curved part (arc), and questions about segments versus sectors.</p> | <p>Say: 'We are going to start our Driving Question Board — this will live on our wall for the whole unit. Every time we learn something new, we will come back and update it.' Distribute sticky notes. 'On your FIRST note, write ONE genuine question you still cannot answer about Mama Wanjiku's watermelon problem. Be specific — not just 'how do you find the area' but WHY does the angle matter? How does the curved rind fit in? On your SECOND note, write one thing you are now more confident about.' Allow 3 minutes of silent writing. Then: 'Come and post your notes.' As learners post, the teacher reads 4–5 questions aloud: 'I see Amina has written: Does the curved part add to the area or is it just the boundary? Excellent — that is going on our DQB. Kevin asks: What if two slices have different radii — can a small angle with a big radius be equal to a big angle with a small radius? Fantastic — that is a deeper question we will return to.' Cluster the questions visibly. Point to the driving question written at the top of the DQB: 'How can we calculate the exact area of any wedge-shaped or segment-shaped piece of a circle — like a watermelon slice at Wakulima</p> | <p>Public Knowledge Tracking via the Driving Question Board. The DQB is a sensemaking artefact that makes the class's collective understanding visible and evolving. By separating 'questions' from 'things we are starting to understand,' the teacher helps learners distinguish between what they know, what they suspect, and what they do not yet know — a metacognitive habit central to the CBE competency of 'learning to learn.'</p> | <p>Teacher reads all DQB notes after class. Key diagnostic indicators: (1) Quality of questions — are they about the mathematical structure (angle, radius, formula) or purely procedural ('how do you solve it')? A class dominated by procedural questions has not yet engaged with the phenomenon at a conceptual level. (2) Quality of 'things we are starting to understand' notes — do learners cite specific evidence from the grid-counting activity? Notes that reference the data ('Sector D had about twice the squares as Sector B because the angle was twice as big') show strong sensemaking; vague notes ('area depends on the circle') show surface engagement. Use this diagnostic to adjust the depth of Lesson 2's instruction.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Market — so that we can solve fair-sharing and design problems in real life?' Say: 'Every question on this board is a step toward answering that driving question. We will not erase a single one — we will answer them one by one.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Arc length as fraction of circumference</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Radius or Diameter of a Circle Given Area (Flexbook)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Learners draw a rough initial model in their books: a circle with a shaded sector, labelled with the two quantities they have identified as relevant (radius <math>r</math> and central angle <math>\theta</math>). They annotate the model with an arrow pointing to the shaded region and write their current best conjecture for what controls its area, in their own words. Below the sketch, they write a 'Model Statement': 'I think the area of a watermelon slice depends on ____ and ____ because ____.' This model will be revised in every subsequent lesson as new mathematical tools are acquired.</p> | <p>Say: 'Scientists and mathematicians build models to capture what they understand so far — even when the model is incomplete. In your book, draw a circle. Shade one sector — one slice. Label the radius <math>r</math> and label the angle in the middle <math>\theta</math> (theta) — that is the Greek letter mathematicians use for angles. Now write your Model Statement: 'I think the area of a watermelon slice depends on ____ and ____ because ____.' You have 3 minutes.' Circulate and read 5–6 model statements. Identify one strong model statement to share (ask permission): 'Zawadi wrote: I think the area depends on the angle and the radius because when the angle doubled in our grid activity, the area doubled, and a bigger radius would make a bigger circle to cut from. That is a strong model. Notice she used EVIDENCE from our activity.' Then say: 'This model is NOT finished — it has no formula yet, no numbers, no proof. That is exactly what this unit is about. Every lesson, we will come back and make this model stronger. By the end of this sub-strand, your model will be able to calculate the EXACT area of any slice to the nearest square centimetre — so every customer at Wakulima Market gets a truly fair share.'</p> | <p>Initial Model Construction with Annotated Conjecture. Asking learners to build an incomplete model deliberately normalises the scientific practice of modelling under uncertainty. The Model Statement template ('depends on ____ and ____ because ____') requires learners to commit to a claim AND a justification, which is the foundation of mathematical reasoning. Revisiting and revising this model across lessons creates a longitudinal sensemaking thread connecting all lessons in the sub-strand.</p> | <p>Teacher collects or photographs 6–8 model statements at the end of class. Readiness indicator for Lesson 2: Does the learner's model statement correctly identify BOTH radius and central angle as the relevant variables, with a reason grounded in the grid-counting evidence? A learner who identifies only angle (but not radius) is partially correct and will need the extension investigation in Lesson 2. A learner who identifies neither variable and relies on visual impression alone may need a bridging conversation at the start of Lesson 2. Record names of learners in each category for differentiated follow-up.</p> |

## D. TEACHER REFLECTION

1. During the Predict phase, what proportion of learners spontaneously identified the central angle as the key variable — and what does this tell you about the prior knowledge your class brings to this sub-strand from their work on circles in Junior School?

2. Were learners' grid-counting estimates accurate enough to show a clear monotonic increase from Sector A to Sector D? If not, what does this suggest about their spatial reasoning skills, and how will you address this in Lesson 2?

3. How rich were the questions learners posted on the Driving Question Board? Did they focus on procedural steps, or did they show curiosity about the mathematical structure (why angle matters, what the curved arc contributes)? What does this tell you about the depth of engagement the phenomenon created?

4. In the Model Building phase, could learners justify their Model Statements with reference to evidence from the activity, or did they rely on intuition alone? What scaffolding will you provide in Lesson 2 to strengthen evidence-based reasoning?

5. Was the Wakulima Market context genuinely engaging for your learners, or did some learners seem disconnected from it? If disconnected, what local adaptation (a different Kenyan market, a different circular food) might create stronger personal relevance for your specific class?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>What did I observe?</b>                   | Learners observed four watermelon-slice sectors — all with radius 7 cm but with central angles of $30^\circ$ , $90^\circ$ , $120^\circ$ , and $180^\circ$ — drawn on centimetre-square grid paper. By counting squares, they found that the estimated area increases as the central angle increases, and that Sector D ( $180^\circ$ ) has approximately twice the area of Sector B ( $90^\circ$ ), mirroring the doubling of the angle.                                                                                                                                                                |
| <b>What did I learn?</b>                     | The area of a sector-shaped piece of a circle is NOT fixed by the radius alone — it changes dramatically with the central angle. When the radius is the same, doubling the central angle approximately doubles the area. Two variables control the size of a watermelon slice: the radius ( $r$ ) and the central angle ( $\theta$ ). Prior methods for finding area (rectangles, triangles) cannot handle the curved boundary of a sector.                                                                                                                                                             |
| <b>How does this explain the phenomenon?</b> | This begins to explain why Mama Wanjiku's customers receive such different amounts of watermelon even though every piece was cut from the same melon: the vendor changed the angle of each cut, and because area is proportional to the central angle, a small change in angle produces a large difference in the amount of fruit. A $30^\circ$ slice contains only one-sixth as much watermelon as the $180^\circ$ slice, even though both pieces look 'like a slice' and share the same radius. The exact formula — which will be derived in Lesson 2 — is needed to calculate fair shares precisely. |

## LESSON 2 (40 minutes): Deriving the Sector Area Formula — From Fraction of a Circle to a Precise Formula

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners derive the formula for the area of a sector by reasoning that a sector is a fraction of a full circle, then apply it to calculate the exact area of watermelon wedges at Wakulima Market.                                                                                                                                                                                                                                                                                                                                   |
| <b>Knowledge</b>            | <ul style="list-style-type: none"> <li>– A sector is a fraction of a full circle determined by the ratio of the central angle to <math>360^\circ</math>.</li> <li>– The area of a sector = <math>(\theta/360^\circ) \times \pi r^2</math>, where <math>\theta</math> is the central angle in degrees and <math>r</math> is the radius.</li> <li>– The two variables that determine sector area are the central angle (<math>\theta</math>) and the radius (<math>r</math>); doubling either changes the area predictably.</li> </ul> |
| <b>Skills</b>               | <ul style="list-style-type: none"> <li>– Derive the sector area formula by reasoning from proportionality between central angle and full circle area.</li> <li>– Calculate the area of a sector given the radius and central angle, expressing answers to 2 decimal places using <math>\pi \approx 3.142</math>.</li> <li>– Verify predictions from Lesson 1 by substituting values into the derived formula.</li> </ul>                                                                                                             |
| <b>Attitudes</b>            | <ul style="list-style-type: none"> <li>– Appreciate that mathematical formulas are not arbitrary rules but logical conclusions derived from reasoning about proportions.</li> <li>– Demonstrate persistence and collaborative spirit when working through derivation steps with peers.</li> </ul>                                                                                                                                                                                                                                    |
| <b>Key Inquiry Question</b> | Where do we use area of part of a circle in real life?                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Purpose in Storyline</b> | In Lesson 1 learners established that sector area depends on both radius and central angle. This lesson advances the storyline by giving learners the precise mathematical tool — the sector area formula — so they can calculate exactly how much watermelon is in each wedge and begin to resolve the fair-sharing dispute at Wakulima Market.                                                                                                                                                                                     |
| <b>Safety Notes</b>         | No specific hazards. Ensure compasses are handled carefully when drawing circles; remind learners to keep compass points directed downward when not in use.                                                                                                                                                                                                                                                                                                                                                                          |

### B. LESSON OVERVIEW

This lesson is the first formal evidence-gathering lesson in the sequence. Building directly on Lesson 1, where learners discovered that sector area is jointly controlled by radius and central angle, learners now derive the sector area formula through proportional reasoning. The central idea is elegantly simple: a sector is nothing more than a fraction of a full circle, and that fraction equals the ratio of the central angle to  $360^\circ$ . By treating the full circle as the reference unit, learners construct the formula themselves rather than receiving it passively.

The lesson opens with learners revisiting their Lesson 1 ideas by making fresh predictions about  $45^\circ$ ,  $90^\circ$ , and  $135^\circ$  watermelon wedges. The teacher uses a large paper circle (a new circle of radius 14 cm) and guides learners to express each sector as a fraction of the whole circle, then multiply by the full circle area ( $\pi r^2$ ). This proportional reasoning — naming the sensemaking strategy as "Fraction of a Whole" — is made visible by folding and overlaying cut-out sectors so learners can physically see that four  $90^\circ$  sectors reconstitute the full circle. Pairs then practise the formula on three worked examples set firmly in the Wakulima Market context, computing exact areas in  $\text{cm}^2$  for wedges of known radii and angles.

By the end of the lesson, learners can calculate the precise area of any sector, compare their results to their Lesson 1 predictions, and articulate why Mama Wanjiku's differently-angled slices are genuinely unequal in area. This resolution of the opening puzzle is recorded on the Driving Question Board and feeds directly into the cumulative model learners are building across the eight-lesson sequence.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                     | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                             | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Area of a sector</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Area of a sector</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Learners individually examine three new sector diagrams ( $45^\circ$ , $90^\circ$ , and $135^\circ$ , all with radius 14 cm drawn on the board). Each learner writes in their exercise book: (a) what fraction of the full circle they think each sector represents, and (b) a numerical estimate of each sector's area in $\text{cm}^2$ , showing their working. After writing, pairs share and compare estimates, noting any disagreements.          | Display the three sector diagrams prominently and say: 'Before we derive any formula today, I want you to think like a mathematician. Look at these three watermelon slices from Mama Wanjiku's stall. In your book, write down: what fraction of the whole watermelon circle is each slice? Then estimate its area.' Allow 3 minutes of silent individual writing. After 3 minutes, say: 'Now turn to your partner and compare your fractions. Do you agree?' Allow 90 seconds of pair discussion. Cold-call 4 learners (select across the room — front-left, back-right, middle, and one who looks uncertain): 'Wanjiku, what fraction did you write for the $90^\circ$ slice?' WAIT TIME: 12 seconds minimum before prompting. Record all four responses on the board under the heading 'Our Predictions — Lesson 2'. Do NOT confirm or reject any prediction yet. Say: 'Hold onto these predictions. By the end of today, you will know exactly who is right — and more importantly, why.' | Anchored Prediction: Learners connect new reasoning (fractions) to the same phenomenon anchor used in Lesson 1. Writing predictions before instruction activates prior knowledge of fractions and primes learners to notice when the formula confirms or corrects their thinking, deepening retention.                                                           | Teacher scans exercise books during pair discussion, looking for: (1) whether learners correctly express $90^\circ$ as $1/4$ of $360^\circ$ — this reveals whether proportional reasoning about angles is accessible; (2) whether area estimates are in a reasonable range (e.g., $90^\circ$ sector of $r=14$ cm should be near $154 \text{ cm}^2$ ). Learners who write 'I don't know how to estimate' are flagged for targeted support during the Observe Phase. |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Area of a sector</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Area of a sector</a>                                                                                                                                                                                   | Learners participate in a structured paper-folding and cut-out investigation. Each pair receives a pre-cut paper circle of radius 14 cm and three pre-cut sector cut-outs: $45^\circ$ , $90^\circ$ , and $135^\circ$ . Learners first fold the $90^\circ$ sector in half and observe it is exactly $1/4$ of the full circle. They then physically tile four $90^\circ$ sectors onto the full circle to confirm $4 \times 90^\circ = 360^\circ$ . Next, | Distribute the pre-cut materials and say: 'Your job is to be a detective. I want you to physically discover how a sector relates to the full circle. Start by placing your $90^\circ$ sector on the full circle. How many times does it fit?' WAIT TIME: 15 seconds. After learners respond, say: 'Write that down as a fraction. Now do the same for the $45^\circ$ and $135^\circ$ sectors — how                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Fraction of a Whole (Concrete-to-Abstract Bridge): Physically tiling sectors onto the full circle makes the proportional relationship tangible before any algebraic formula is introduced. This concrete manipulation prevents the formula from being received as an arbitrary rule and instead positions it as a description of something learners have already | Teacher checks the observation table during circulation. Key indicators: (1) Learners correctly write $\theta/360^\circ$ for all three sectors (not $\theta/180^\circ$ or $\theta/\pi$ — common errors); (2) Learners can explain in words why the denominator is $360^\circ$ ; (3) Learners who tile sectors correctly but cannot express it as a fraction need targeted support linking the physical action to the                                               |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>learners complete a guided observation table in their books with three columns: Central Angle (<math>\theta</math>), Fraction of Full Circle (<math>\theta/360^\circ</math>), and 'What we multiplied the full circle area by'. They record observations for all three sectors. Finally, the teacher reveals the full circle area formula (<math>A = \pi r^2</math>) and learners calculate the full circle area for <math>r = 14</math> cm as a shared step.</p>                                                                                                                                                                                                                                                                                                                                                                                        | <p>many of each would fill the full circle?' Circulate actively, pausing at each pair for 20–30 seconds. Use probing questions: 'How did you get that fraction?', 'What does the 360 in the denominator represent?' After 6 minutes of hands-on work, bring the class together. Draw a table on the board and cold-call 3 learners to fill in the three rows of the fraction column. Then say: 'The full circle has area <math>\pi r^2</math>. For our watermelon with <math>r = 14</math> cm, let's calculate that together.' Write on the board:<br/> <math>A_{\text{circle}} = \pi \times 14^2 = \pi \times 196 \approx 3.142 \times 196 = 615.83 \text{ cm}^2</math>. Say: 'Now — if the full circle is 615.83 <math>\text{cm}^2</math>, and a <math>90^\circ</math> sector is one-quarter of the circle, what is the sector's area?' WAIT TIME: 12 seconds. Cold-call 1 learner. Confirm the answer (<math>\approx 153.96 \text{ cm}^2</math>) and connect it back to the prediction board: 'Look at your prediction from the start of the lesson. How close were you?'</p> | <p>seen and touched — directly analogous to Mama Wanjiku's physical cutting of the watermelon.</p>                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>symbolic fraction. Record names of learners who are consistently inaccurate for follow-up in the Explain Phase.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Area of a circle</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of a circle</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar</a></p> | <p>Learners formalise the derivation in their books. Working individually, each learner writes out the full derivation in three steps: Step 1 — express the sector as a fraction of the circle (<math>\theta/360^\circ</math>); Step 2 — the full circle area is <math>\pi r^2</math>; Step 3 — therefore, Area of sector = <math>(\theta/360^\circ) \times \pi r^2</math>. Learners then apply the formula to three graded practice problems, all set in the Wakulima Market context: (A) a <math>60^\circ</math> slice of radius 21 cm (a mango segment at a nearby stall), (B) a <math>120^\circ</math> slice of radius 10.5 cm (a pawpaw wedge), and (C) Mama Wanjiku's three watermelon slices — <math>45^\circ</math>, <math>90^\circ</math>, and <math>135^\circ</math> at <math>r = 14</math> cm — for which they now calculate exact areas and</p> | <p>Say: 'We are now going to capture what we discovered as a formula. Write this heading in your book: Derivation of the Sector Area Formula.' Guide learners step-by-step, writing each line on the board and pausing for learners to copy and annotate: 'A sector with central angle <math>\theta</math> is (<math>\theta/360^\circ</math>) of the full circle. The full circle has area <math>\pi r^2</math>. Therefore: Area of sector = <math>(\theta/360^\circ) \times \pi r^2</math>.' Say: 'Box this formula. Every time you use it, you should be able to explain where it came from — from the paper circle we just folded.' Assign the three practice problems. Say: 'Work individually and silently for 5 minutes. Show all three steps for each problem.' After 5 minutes,</p>                                                                                                                                                                                                                                                                                      | <p>Guided Derivation with Annotated Formula: Rather than stating the formula and practising it, learners write the derivation in their own books with annotations explaining each line. This metacognitive narration strategy (writing 'because the sector is a fraction of the whole circle' next to the formula) ensures the formula is stored with meaning attached, not as an isolated rule. Connecting back to the three specific Wakulima Market slices grounds abstract algebra in the concrete phenomenon.</p> | <p>Teacher collects answers to Problem C as a key formative checkpoint. Correct answers: <math>45^\circ</math> sector <math>\approx 76.99 \text{ cm}^2</math>; <math>90^\circ</math> sector <math>\approx 153.99 \text{ cm}^2</math>; <math>135^\circ</math> sector <math>\approx 230.98 \text{ cm}^2</math>. Learners who obtain these answers (<math>\pm 0.5 \text{ cm}^2</math> rounding tolerance) have met the lesson SLO. Common errors to watch for: (1) using diameter instead of radius; (2) writing <math>(\theta/180^\circ)</math> instead of <math>(\theta/360^\circ)</math>; (3) forgetting to square the radius. Any learner making error (2) or (3) on more than one problem is flagged for re-teaching before Lesson 3.</p> |

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| <a href="#">readings</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>compare with their Lesson 2 predictions.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>cold-call 3 different learners (one per problem) to write their solutions on the board. For each solution ask the class: 'Do you agree with this answer? Give a thumbs up if you got the same, thumbs sideways if slightly different, thumbs down if very different.' Address any thumbs-down responses immediately with targeted re-teaching. For Problem C, say: 'Now compare your exact answers with the predictions you wrote at the start of today's lesson. What do you notice? Has anything surprised you?' WAIT TIME: 12 seconds. Cold-call 2 learners.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of a circle intuition</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Radians &amp; arc length</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to the class Driving Question Board (established in Lesson 1). Each learner writes one sticky note (or index card) responding to the prompt: 'What can I now calculate that I couldn't calculate before this lesson?' They also annotate the three prediction cards from this lesson's Predict phase with the actual computed values (in a different colour) so the board shows both prediction and evidence side by side. Two volunteers read their sticky notes aloud before they are posted.</p> | <p>Direct learners' attention to the DQB: 'Look at the predictions we recorded at the start of today's lesson. We now have exact numbers. Take your index card and write: what new calculating power do you have today that you didn't have after Lesson 1?' Allow 2 minutes of individual writing. Invite 2 volunteers to read their cards aloud before posting. Then point to the driving question displayed at the top of the DQB: 'How can we calculate the exact area of any wedge-shaped piece of a circle so that every customer at Wakulima Market gets a fair share?' Say: 'Can we answer this question for sectors now?' Elicit a brief class response. Then say: 'But look — Mama Wanjiku's watermelon also has a curved rind and some slices are cut all the way through. What if the piece is not a full wedge but a shape bounded by a chord rather than two radii? Write one question on the board that this new knowledge makes you wonder about.' Collect 2–3 student-generated questions and</p> | <p>Evidence Layering on the DQB: By writing actual computed values next to earlier predictions in a contrasting colour, the DQB becomes a visible record of the class's growing precision. The shift from 'I estimated about 150 cm<sup>2</sup>' to 'I can now calculate exactly 153.99 cm<sup>2</sup>' is captured publicly, reinforcing the epistemic value of deriving and applying a formula. Student-generated questions on the board sustain curiosity and create authentic need for the next lesson.</p> | <p>Teacher reads the sticky notes as they are posted. A well-formed response names the formula (e.g., 'I can now use <math>\text{Area} = (\theta/360^\circ) \times \pi r^2</math> to calculate the exact area of any sector') and connects it to the phenomenon (e.g., 'so I can tell Mama Wanjiku's customers exactly how many cm<sup>2</sup> are in their slice'). Vague responses such as 'I learned about circles' indicate surface-level engagement and require a follow-up oral probe: 'Can you show me with an example?'</p> |

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| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Arc length as fraction of circumference</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Arc length</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners update their cumulative visual model of a watermelon cross-section, which was started in Lesson 1. In Lesson 1, the model showed a circle with a shaded sector labelled with <math>r</math> and <math>\theta</math> and arrows indicating 'bigger angle = bigger area'. In this lesson, learners add: (1) the sector area formula written inside one of the sectors; (2) exact area values (in <math>\text{cm}^2</math>) labelled on each of the three sectors (<math>45^\circ \approx 76.99 \text{ cm}^2</math>, <math>90^\circ \approx 153.99 \text{ cm}^2</math>, <math>135^\circ \approx 230.98 \text{ cm}^2</math>); (3) a ratio note showing that the <math>135^\circ</math> slice is exactly <math>3\times</math> the area of the <math>45^\circ</math> slice, reinforcing proportionality. Learners write one sentence beneath the model: 'Mama Wanjiku's slices are NOT equal — the <math>135^\circ</math> slice has 3 times more watermelon than the <math>45^\circ</math> slice.'</p> | <p>Say: 'Open to your model from Lesson 1. Today we are going to make it more precise. Add the formula, the exact areas, and the ratio relationship we discovered.' Display a reference diagram on the board showing where each annotation should go. Circulate for 3 minutes, checking that: (1) the formula is written correctly (not from memory error); (2) the ratio <math>3:1</math> is correctly identified (<math>135^\circ/45^\circ = 3</math>). As learners finish, cold-call 1 learner to hold up their model and explain one annotation to the class in one sentence. Say: 'Your model is now a mathematical argument. If Mama Wanjiku's customer challenged you, you could use this diagram to prove that the <math>135^\circ</math> slice is three times larger than the <math>45^\circ</math> slice.' Finally, preview the next lesson: 'Next time, we will look at a different kind of slice — one where the cut is straight across, like a chord, not two radii. This is called a segment. Does your formula still work? That is our next puzzle.'</p> | <p>Cumulative Model Revision: The model serves as a running scientific and mathematical argument that grows in precision across lessons. Each revision adds a layer of evidence (qualitative in Lesson 1, quantitative in Lesson 2) without erasing earlier thinking, so learners can trace their own conceptual development. Labelling exact values and ratios directly on the watermelon cross-section keeps the abstract formula anchored to the real-world phenomenon.</p> | <p>Teacher checks models for three specific features: (1) formula written correctly as <math>(\theta/360^\circ) \times \pi r^2</math>; (2) at least two of the three sector areas labelled correctly (within <math>\pm 1 \text{ cm}^2</math> rounding); (3) a sentence or annotation that correctly connects formula output to the fairness question. Models missing the ratio annotation are flagged — this indicates the learner can compute but has not yet connected calculation to proportional reasoning, which is needed for Lesson 3.</p> |

## D. TEACHER REFLECTION

1. During the Observe Phase, did all pairs correctly tile the  $90^\circ$  sector four times onto the full circle and express this as the fraction  $1/4$ ? If several pairs wrote  $1/2$  or  $2/4$  without simplifying, how will I address fraction simplification before Lesson 3 without derailing the sector content?
2. In the Explain Phase, which of the three common errors (using diameter, using  $180^\circ$  denominator, or forgetting to square  $r$ ) appeared most frequently in learners' Problem C solutions? What targeted re-teaching micro-moment can I build into the opening 3 minutes of Lesson 3?
3. Did the sticky notes on the DQB show genuine connection to the phenomenon (mentioning Mama Wanjiku, Wakulima Market, fair shares), or were they generic mathematical statements? How can I use the phenomenon more explicitly in my questioning next lesson?
4. Were the student-generated questions posted at the end of the DQB phase pointing naturally toward segments, or did learners raise other ideas (e.g., arc length, volume of the watermelon)? How will I honour and sequence those questions in the remaining lessons?
5. When cold-calling during the Predict Phase, did I genuinely wait at least 12 seconds, or did I prompt too early? Review my timing — was there equitable cold-calling

across gender and seating zones in the classroom?

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| What did I observe?                                          | Learners physically tiled pre-cut sector cut-outs onto a full paper circle of radius 14 cm and observed that four 90° sectors, eight 45° sectors, or approximately 2.67 of the 135° sectors reconstitute the full circle — confirming that each sector is a measurable fraction ( $\theta/360^\circ$ ) of the whole.                                                                                                                                                                             |
| What did I learn?                                            | The area of a sector equals the fraction of the full circle it occupies multiplied by the full circle area: $\text{Area} = (\theta/360^\circ) \times \pi r^2$ . For Mama Wanjiku's watermelon ( $r = 14$ cm), the 45° slice $\approx 76.99$ cm <sup>2</sup> , the 90° slice $\approx 153.99$ cm <sup>2</sup> , and the 135° slice $\approx 230.98$ cm <sup>2</sup> — confirming that the 135° slice is exactly three times larger than the 45° slice.                                            |
| How does this explain the phenomenon?                        | The fair-sharing dispute at Wakulima Market can now be resolved mathematically: Mama Wanjiku's slices are provably unequal because they have different central angles. The sector area formula gives every customer an objective tool to verify the size of their slice in cm <sup>2</sup> , directly answering the driving question for wedge-shaped pieces. The remaining puzzle is whether the same approach works for a segment (a slice bounded by a straight chord rather than two radii). |

## LESSON 3 (80 minutes (2 × 40-minute lessons)): Investigating Sector Area with Cut-Outs

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>Purpose</b>              | Learners derive and apply the formula for the area of a sector by physically constructing, measuring, and comparing paper cut-out sectors of different radii and angles — directly connecting their findings to the watermelon-slice phenomenon at Wakulima Market.                                                                                                                                                                                                                                |
| <b>Knowledge</b>            | Learners determine that the area of a sector of a circle is given by $A = (\theta/360) \times \pi r^2$ , where $\theta$ is the angle subtended at the centre and $r$ is the radius, and that changing either variable changes the area proportionally.                                                                                                                                                                                                                                             |
| <b>Skills</b>               | Learners measure radius and central angle of paper cut-out sectors, calculate sector areas using the derived formula, compare results across different sectors, and communicate mathematical reasoning to peers.                                                                                                                                                                                                                                                                                   |
| <b>Attitudes</b>            | Learners appreciate that precise mathematical formulas enable fair and accurate measurement of curved shapes, and demonstrate Unity and Peace by collaborating respectfully when constructing cut-outs and sharing findings.                                                                                                                                                                                                                                                                       |
| <b>Key Inquiry Question</b> | Where do we use area of part of a circle in real life?                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Purpose in Storyline</b> | In Lesson 2 learners derived the sector-area formula from fractions of a circle. In Lesson 3 they test and confirm it with their own data: they construct sector cut-outs, measure them, and check the formula against the pattern in their results. This gives them the first precise mathematical tool for answering the driving question — calculating exactly how many cm <sup>2</sup> of watermelon are in any given wedge, so that Mama Wanjiku at Wakulima Market can price and cut fairly. |
| <b>Safety Notes</b>         | Learners use scissors and compasses during cut-out construction. Remind learners to handle scissors with blades pointing downward and to place compasses on the desk when not in use. No running with sharp instruments.                                                                                                                                                                                                                                                                           |

### B. LESSON OVERVIEW

Learners construct paper cut-outs of sectors with varying central angles (30°, 60°, 90°, 120°, 180°) and a fixed radius, then with varying radii and a fixed angle. They measure, record, and calculate areas, ultimately confirming  $A = (\theta/360) \times \pi r^2$  (derived in Lesson 2) from the pattern in their data. They then test the formula against the watermelon-slice phenomenon by calculating the area of two specific vendor cuts described in the anchoring scenario and determining whether the vendor's cuts are truly fair. The lesson builds on the formula derived in Lesson 2 and supplies the quantitative toolkit needed for Lessons 4–8.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                 | Learner Experience                                                                                                                                                                                                                         | Teacher Moves                                                                                                                                                                                                                      | Sensemaking Strategy                                                                                                                                                                                                            | Formative Assessment Strategy                                                                                                                                                                                             |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Area of a sector</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a> | Each learner receives a half-page printed sheet showing five sector outlines: all with radius 7 cm but central angles of 30°, 60°, 90°, 120°, and 180°. Without any formula, learners individually write two predictions in their exercise | Distribute the printed sector-outline sheet. Say: 'Before we calculate anything today, I want you to THINK about what you already believe. Look at these five watermelon slices — same melon, same radius, different cuts. In your | Predict-then-Test anchoring: By committing predictions to writing before any instruction, learners activate prior knowledge about proportionality and area, create cognitive dissonance that motivates the activity, and have a | Circulate during silent writing. Look for: (a) learners who correctly predict a proportional (linear) relationship between angle and area — this shows strong proportional reasoning; (b) learners who predict the radius |

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| <p> <b>EXERCISE:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar readings</p>                                                                                                                                                                                                                                                                                                                                                                           | <p>books: (1) 'I predict the sector with angle <math>180^\circ</math> has ____ times the area of the sector with angle <math>90^\circ</math>, because...' and (2) 'If I double the radius but keep the angle fixed, I predict the area will ____.' Learners then compare predictions with a desk partner for 2 minutes before a whole-class share-out.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>book, complete both prediction sentences. You have 3 minutes of silent thinking time — GO.' Set a visible 3-minute timer. After 3 minutes say: 'Turn to your desk partner — compare your predictions. Do you agree? If not, why not? You have 2 minutes.' After partner talk, cold-call 3 learners (one who predicted a doubling relationship, one who did not, one who is unsure) using: 'Kamau — share your prediction for question 1 and your reasoning.' Apply 10-second wait time after each cold-call before accepting any response. Record two or three contrasting predictions visibly on the board under the heading 'OUR PREDICTIONS — we will test these today.'</p>                                                                                                                                                                                                                                                                                   | <p>personal benchmark against which to evaluate their later findings. This directly mirrors the market vendor's implicit prediction that equal angles mean equal area.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>relationship will also be linear (rather than quadratic) — this is the key misconception to address in Phase 3. Note names of both groups for targeted questioning in Phase 3. Observable indicator: every learner has at least one written prediction sentence before partner talk begins.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <p><b>Observe Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>Groups of 4 receive a Materials Pack: one sheet of A4 card, a compass, protractor, ruler, scissors, and a recording table (pre-printed or copied from the board). PART A — Fixed radius, varying angle (15 min): Each group draws and cuts out 5 sectors on card, all with radius 7 cm, at angles <math>30^\circ</math>, <math>60^\circ</math>, <math>90^\circ</math>, <math>120^\circ</math>, <math>180^\circ</math>. For each sector they record the central angle <math>\theta</math> and radius <math>r</math>, and calculate the fraction <math>\theta/360</math>. They record these in their table. PART B — Fixed angle, varying radius (8 min): Groups draw and cut out 3 sectors all at <math>60^\circ</math>, with radii 4 cm, 7 cm, and 10 cm. They record <math>r</math>, <math>r^2</math>, and the fraction <math>\theta/360</math> for each. PART C — Pattern hunt (5 min): Groups inspect their two tables and write one sentence describing the pattern they see between <math>(\theta/360)</math> and area, and another sentence describing the relationship between <math>r</math> and area.</p> | <p>Before groups begin say: 'You are now Mama Wanjiku's quality-control team. She needs to know the exact area of each wedge — not an estimate, the EXACT area. Your cut-outs are your evidence. Every measurement must be recorded in your table — a scientist at KEMRI would not skip a row.' Circulate continuously. At the 7-minute mark, pause the class for a 60-second mid-activity check: cold-call one group — 'Group 3 — Aisha, tell me: what is <math>\theta/360</math> for your <math>90^\circ</math> sector?' (10-second wait time). If the group says 0.25, affirm and say: 'Hold that number — it will matter in Phase 3.' If incorrect, prompt: 'What fraction of the full <math>360^\circ</math> is <math>90^\circ</math>?' For Part B, prompt groups who finish early: 'Look at your <math>r</math> column and your area column. What happens to the area when <math>r</math> doubles from 4 to 8? Is that what you predicted?' During Part C,</p> | <p>Data-Table Pattern Recognition with Physical Manipulatives: Learners generate their own data through measurement of self-constructed objects, rather than reading values from a textbook. The physical act of cutting sectors at different angles makes the proportional relationship between angle and area visceral and memorable. Separating the two variables (Part A then Part B) follows a controlled-variable investigation structure (NGSS Science and Engineering Practice 3: Planning and Carrying Out Investigations), enabling learners to isolate the effect of each variable on area.</p> | <p>Collect observable evidence at three checkpoints: (1) At 10 minutes — every group should have at least 3 completed rows in their Part A table with <math>\theta/360</math> correctly calculated; (2) At 20 minutes — Part B table should show <math>r^2</math> values calculated, not just <math>r</math>; (3) At 28 minutes — each group's Part C pattern sentence should mention both angle and radius. Key misconception to watch: groups who calculate <math>\theta/360 \times r</math> (instead of <math>\theta/360 \times r^2</math>) in Part C — flag these groups for direct questioning in Phase 3. Indicator of success: 100% of groups have a completed two-part data table with at least one pattern sentence written.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | circulate and listen for the word 'proportional' — if no group uses it, plant the seed: 'Is the relationship between $\theta$ and area the same KIND of relationship as between $r$ and area? Discuss.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of a circle</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of a circle</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Whole-class discussion leads to derivation of the sector area formula. Learners then apply the formula to two specific cuts: Cut A (<math>\theta = 90^\circ</math>, <math>r = 7</math> cm) and Cut B (<math>\theta = 180^\circ</math>, <math>r = 7</math> cm) — the two wedges the Wakulima Market vendor claimed were 'fair.' Learners calculate both areas, compare them, and write a one-sentence verdict: 'The vendor's cuts ARE / ARE NOT fair because...' They also revisit their Phase 1 predictions and annotate them with 'CONFIRMED' or 'REVISED — because...'</p> | <p>Open with: 'Let's confirm the formula from Lesson 2 using YOUR data. Which formula did we derive, and does your data support it?' Draw a blank formula template on the board: Area of sector = <math>\frac{\theta}{360} \times \text{Area of full circle}</math>. Ask cold-call question (10-second wait): 'Juma — from your Part A table, what fraction of the full circle is a <math>60^\circ</math> sector?' After response, fill in: 'So Area of <math>60^\circ</math> sector = <math>\frac{60}{360} \times \pi r^2</math>.' Ask: 'Wanjiku — generalise that. If the angle is <math>\theta</math>, what is the formula?' (10-second wait). Write <math>A = \frac{\theta}{360} \times \pi r^2</math> on the board. Then ask: 'From your Part B data — when you DOUBLED the radius, what happened to the area?' Cold-call a third learner (10-second wait). Establish that area quadruples, confirming the <math>r^2</math> term. Now pivot to the phenomenon: 'Mama Wanjiku cuts two slices from a melon of radius 7 cm: one at <math>90^\circ</math> and one at <math>180^\circ</math> (pieces B and D from Lesson 1). Use the formula — calculate both areas in your book. You have 4 minutes.' After 4 minutes, cold-call two learners to share answers and write them on the board. Ask: 'Is that a fair price if she charges the same for both? What would YOU tell her?' End Phase 3 by directing learners back to their Phase 1 prediction page: 'Annotate your predictions — were you right? Write CONFIRMED or REVISED and explain why in one sentence.'</p> | <p>Guided Derivation with Phenomenon Application (NGSS SEP 6 — Constructing Explanations): Rather than presenting the formula as a given fact, the teacher scaffolds learners to construct it from their own data, then immediately apply it to the anchoring phenomenon. This two-step move — derive, then deploy — ensures the formula is understood as an explanation of the physical reality of the watermelon slices, not an abstract rule. Connecting back to the vendor scenario embeds the mathematics in a justice and fairness context meaningful to Kenyan learners.</p> | <p>Collect exercise books during the 4-minute calculation window and scan 4–5 books for: (a) correct substitution of <math>\theta = 90</math> and <math>r = 7</math> into <math>A = \frac{\theta}{360} \times \pi r^2</math>; (b) correct arithmetic (area of Cut A <math>\approx 38.48</math> cm<sup>2</sup>; area of Cut B <math>\approx 76.97</math> cm<sup>2</sup>); (c) a written verdict sentence. Misconception alert: learners who write <math>A = \frac{\theta}{360} \times 2\pi r</math> (using circumference formula instead of area formula) — address immediately by asking: 'Is <math>\pi r^2</math> the area or the circumference of the full circle?' Observable indicator of meeting expectations: at least 80% of learners produce a correct numerical answer for both cuts and a verdict sentence before the phase ends.</p> |

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| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Arc length as fraction of circumference</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Radians &amp; arc length</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Each learner receives two sticky notes. On the GREEN note they write one thing they can NOW answer about the driving question that they could not answer before this lesson (e.g., 'I can now calculate the exact area of any wedge if I know <math>r</math> and <math>\theta</math>'). On the ORANGE note they write one new question this lesson has raised (e.g., 'What if the watermelon slice is not a full wedge — what if it has a flat chord cut across the bottom?'). Learners post notes on the class DQB, which is organised into columns: WHAT WE PREDICTED, WHAT WE OBSERVED, WHAT WE CAN NOW EXPLAIN, WHAT WE STILL WONDER. The teacher reads three orange notes aloud and tells the class which upcoming lesson will address each question.</p>                                                                                       | <p>Say: 'Our Driving Question is: how can we calculate the exact area of any wedge or segment-shaped piece of a circle? Look at the DQB. In Lesson 2 we filled the column with the formula we derived. Today we fill WHAT WE CAN NOW EXPLAIN.' Hand out sticky notes. Allow 3 minutes of silent writing. As learners post notes, read three orange notes aloud — select at least one that anticipates the segment (chord-cut) concept. Say: 'This question — about a flat-bottom slice — is EXACTLY what Lesson 5 will answer. Keep wondering.' Point to the OBSERVED column from Lesson 2 and say: 'Notice how our observation that a bigger angle means a bigger piece is now backed by a formula. That is the power of mathematics — we moved from seeing to calculating.'</p>           | <p>Driving Question Board as Cumulative Sense-Making Record (NGSS SEP 8 — Obtaining, Evaluating, and Communicating Information): The DQB makes the progression of understanding visible across the eight-lesson sequence. Separating 'can now explain' from 'still wonder' trains learners to distinguish between resolved and unresolved questions — a core scientific literacy skill. The orange notes also provide the teacher with formative data on which concepts need pre-teaching before Lesson 5.</p>                                       | <p>Collect and read all orange sticky notes after class. Categorise them into: (a) segment vs. sector confusion — indicates readiness for Lesson 5; (b) radius-effect questions — indicates the <math>r^2</math> relationship needs reinforcement in Lesson 4; (c) real-life application questions — indicates high engagement and readiness for design tasks in Lesson 7. Observable indicator: every learner posts both a green and an orange note.</p>                                                                                                                                                                                                                                |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Impact of increasing the radius</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Arc length</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan</p>                                                                                                                                                                                                             | <p>Learners return to their cumulative annotated diagram — a labelled drawing of a circular watermelon cross-section they began in Lesson 1 and annotated in Lesson 2. Today they add three new elements: (1) they label a sector with the formula <math>A = (\theta/360) \times \pi r^2</math> written directly on the wedge; (2) they annotate one wedge with a worked numerical example using the Wakulima Market values (<math>\theta = 90^\circ</math>, <math>r = 7\text{ cm} \rightarrow A \approx 38.48\text{ cm}^2</math>); (3) they draw an arrow from the formula to a speech bubble containing the text: 'This is why the vendor's <math>90^\circ</math> and <math>180^\circ</math> slices are NOT equal even though the radius is the same.' Learners write a one-line 'Model Update Statement' at the bottom of their diagram: 'I used</p> | <p>Say: 'Get out your cumulative watermelon diagram. You have been building this model since Lesson 1. Today you have a new, powerful piece of evidence to add — the formula. Add all three annotations I have written on the board.' Write the three annotation instructions on the board for reference. Circulate and look for learners who write the formula incorrectly (e.g., missing the <math>r^2</math> or writing <math>r</math> instead of <math>r^2</math>). Ask targeted questions: 'Chebet — in your formula, why is it <math>r</math>-squared and not just <math>r</math>? What did your Part B data tell you?' (10-second wait). In the final 2 minutes, cold-call one learner to read their Model Update Statement aloud. Ask the class: 'Does everyone agree with that</p> | <p>Cumulative Model Revision (NGSS SEP 2 — Developing and Using Models): The annotated diagram serves as a running scientific model that learners revise with each new lesson. Adding the formula directly onto the watermelon diagram — rather than into a separate notes section — reinforces that the formula is an explanation of the physical phenomenon, not an abstract mathematical object. The 'I used to think / Now I think' frame (adapted from Project Zero's Thinking Routine) makes conceptual change explicit and self-directed.</p> | <p>At the end of class, collect 8 cumulative diagrams (2 per group) for a rapid check. Look for: (a) formula written correctly with both <math>\theta</math> and <math>r^2</math> present; (b) a worked numerical example with correct substitution; (c) a Model Update Statement that references BOTH variables. Learners whose diagrams are missing any element receive a written prompt sticker: 'What does your Part B data tell you about what happens to area when <math>r</math> changes?' to guide self-correction before Lesson 4. Observable indicator of meeting expectations: formula is correctly written on the diagram with at least one annotated numerical example.</p> |

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| Academy (English - US curriculum)<br><a href="#">Search ARES for similar readings</a> | to think the size of a wedge depended only on how it looked. Now I think it depends on BOTH the angle $\theta$ and the radius $r$ , because $A = (\theta/360) \times \pi r^2$ .' | statement? Give me a thumbs up if your statement says something similar.' Use thumbs to gauge class-wide consolidation before dismissal. |  |  |
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### D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. **FORMULA DERIVATION:** Did learners genuinely derive  $A = (\theta/360) \times \pi r^2$  from their own data, or did they copy it from the board? In your next lesson, ask two or three learners to explain in their own words why the formula contains  $r^2$  — their explanations will reveal the depth of conceptual understanding.
2. **THE  $r^2$  MISCONCEPTION:** How many learners initially wrote  $r$  instead of  $r^2$  in their formula? This is the most common error in this sub-strand. If more than 30% of the class made this error, open Lesson 4 with a brief re-engagement with the Part B data: 'When Amina's radius went from 4 cm to 8 cm (doubled), what happened to the area?' Use their own recorded data to self-correct.
3. **FAIR-SHARING VERDICT:** Did learners engage with the justice dimension of the vendor scenario — that unequal cut angles produce unfair portions? Or did they treat it as a purely computational exercise? If the fairness angle was missed, find a moment in Lesson 4 to ask: 'What would you charge per  $\text{cm}^2$  for a watermelon slice at Wakulima Market if the whole melon costs KSh 150?'
4. **DQB ORANGE NOTES:** Review the orange sticky notes tonight. If most questions are about segments (flat-bottom slices), learners are ahead of pace and you can briefly preview the chord-cut concept at the start of Lesson 4. If questions are still about sectors, consolidate before moving forward.
5. **GROUP DYNAMICS:** Did all learners in each group of 4 physically handle the compass and scissors, or did one learner dominate the construction? For Lesson 4, assign roles explicitly: Compass Operator, Protractor Reader, Recorder, and Reporter — rotating roles ensures equitable participation and richer individual learning.

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>What did I observe?</b>                   | Learners constructed paper cut-out sectors of a circle with varying central angles ( $30^\circ$ , $60^\circ$ , $90^\circ$ , $120^\circ$ , $180^\circ$ ) at a fixed radius of 7 cm, and sectors at a fixed angle of $60^\circ$ with varying radii (4 cm, 7 cm, 10 cm). They measured, recorded data in tables, and identified patterns — that area scales linearly with central angle and as the square of the radius.                                                              |
| <b>What did I learn?</b>                     | The area of a sector of a circle is given by $A = (\theta/360) \times \pi r^2$ , where $\theta$ is the central angle in degrees and $r$ is the radius. Doubling the angle doubles the area; doubling the radius quadruples the area. Learners applied this formula to calculate the areas of two Wakulima Market watermelon wedges ( $\theta = 90^\circ$ and $\theta = 180^\circ$ , $r = 7$ cm) and determined the vendor's equal-price claim was mathematically unfair.           |
| <b>How does this explain the phenomenon?</b> | The sector area formula (derived in Lesson 2) comes from the fact that a sector is a fraction ( $\theta/360$ ) of the full circle. Because the full circle has area $\pi r^2$ , any sector has area $(\theta/360) \times \pi r^2$ . This explains why the vendor's two wedge-cuts at different angles produce measurably different areas even though both come from the same melon — changing only the angle of the cut directly and proportionally changes the area of the piece. |

## LESSON 4 (40 minutes): Cutting Watermelon Fairly: Applying the Area of a Sector Formula

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>Purpose</b>              | Learners consolidate and apply the formula for the area of a sector by physically cutting out sectors of different angles from printed circles, tabulating results, and identifying the proportional relationship between angle size and area.                                                                                                                                                                                                                                                                   |
| <b>Knowledge</b>            | Learners consolidate that the area of a sector of a circle is a fraction of the total circle area, given by $\text{Area} = (\theta/360) \times \pi r^2$ , where $\theta$ is the central angle in degrees and $r$ is the radius.                                                                                                                                                                                                                                                                                  |
| <b>Skills</b>               | Learners measure angles and radii using a protractor and ruler, calculate sector areas as fractions of the full circle area, tabulate data, and identify a mathematical pattern from evidence.                                                                                                                                                                                                                                                                                                                   |
| <b>Attitudes</b>            | Learners appreciate that mathematics provides an exact, fair method for comparing and calculating wedge-shaped portions — resolving real disputes like those at Wakulima Market — and value precision and collaborative inquiry.                                                                                                                                                                                                                                                                                 |
| <b>Key Inquiry Question</b> | Where do we use area of part of a circle in real life?                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Purpose in Storyline</b> | In Lessons 1–3, learners established that the angle of a wedge cut changes the size of a watermelon slice, derived the sector formula (Lesson 2) and confirmed it with cut-outs (Lesson 3). In Lesson 4 learners consolidate: they cut sectors of five angles, compute each area as a fraction of $\pi r^2$ , and verify the formula $\text{Area} = (\theta/360) \times \pi r^2$ . This formula is the quantitative tool needed to answer the driving question and resolve the Wakulima Market fairness dispute. |
| <b>Safety Notes</b>         | Learners will use scissors to cut printed circles. Remind all learners to cut away from their bodies and neighbouring learners. Collect and store scissors immediately after the cutting phase. Learners with any hand injury should measure and record rather than cut.                                                                                                                                                                                                                                         |

### B. LESSON OVERVIEW

Learners work in groups of 4. Each group receives a printed circle of radius 7 cm, a protractor, scissors, a ruler, and a recording table. Groups mark and cut out sectors of 30°, 60°, 90°, 120°, and 180° from their circles. They arrange the sectors side by side, observe the dramatic size differences, measure each sector's angle and radius, then calculate the area of each sector as a fraction of the full circle area ( $\pi r^2 = 154 \text{ cm}^2$ ). Results are tabulated across the whole class and the formula from Lesson 2 —  $\text{Area of sector} = (\theta/360) \times \pi r^2$  — is confirmed through structured questioning. The lesson ends with groups writing the formula in their own words and applying it to two new sector problems set in a Kenyan context (a slice of ugali cake and a door-sweep in Kisumu). Every step is anchored to the Wakulima Market watermelon phenomenon: learners are calculating exactly how much watermelon is in each wedge.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                           | Learner Experience                                                                                                                                            | Teacher Moves                                                                                                                                                                   | Sensemaking Strategy                                                                                                                                         | Formative Assessment Strategy                                                                                                                                              |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Area of a sector</a><br>Source: Khan Academy (English) | Each group is shown a circle drawn on the board (radius 7 cm) with five sectors marked: 30°, 60°, 90°, 120°, 180°. Without calculating, learners individually | Draw the five-sector circle on the board clearly, labelling each angle. Say: 'Before we cut anything, I want your best guess. Look at the full circle — its area is 154 cm². In | Predict-before-observe (anticipation guide): By committing to a numerical estimate before gathering data, learners activate prior knowledge of fractions and | Listen for whether learners use proportional reasoning (e.g., '90° is a quarter of 360° so the area should be a quarter of 154') versus additive or visual guessing. Cold- |

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| <p>- US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                  | <p>write two predictions in their exercise books: (1) 'If the full circle area is <math>154 \text{ cm}^2</math>, what do you estimate the area of the <math>90^\circ</math> sector is?' and (2) 'Which two sectors together would give exactly half the circle?' Learners share predictions with their group partner for 1 minute, then the teacher cold-calls 3 learners to share with the class.</p>                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>your exercise book, write your estimate for the area of the <math>90^\circ</math> sector and explain your reasoning.' WAIT TIME: 30 seconds of silent thinking. Then say: 'Share your prediction with your partner. You have 1 minute.' After 1 minute, cold-call 3 learners by name: 'Amina, what did you predict and why? ... Brian, do you agree with Amina? ... Zawadi, what was different about your thinking?' Record the range of predictions on the board without correcting them — label it 'OUR PREDICTIONS'. Say: 'We will come back to check these predictions against our evidence.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>circle area, and create a cognitive need to test their predictions. Recording predictions publicly creates accountability and makes conceptual change visible later.</p>                                                                                                                                                                                                                                                 | <p>call responses reveal whether the fraction-of-a-whole concept is accessible. If fewer than 2 of 3 cold-called learners use any fractional reasoning, pause and ask the whole class: 'What fraction of a full turn is <math>90^\circ</math>?' before proceeding.</p>                                                                                                                                                                                                                                                                                                                                                                          |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Each group receives: one printed circle (radius = 7 cm, with <math>30^\circ</math>, <math>60^\circ</math>, <math>90^\circ</math>, <math>120^\circ</math>, <math>180^\circ</math> sectors pre-marked), scissors, a protractor, a ruler, and a group recording table (columns: Angle <math>\theta</math>, Radius <math>r</math>, Full circle area <math>\pi r^2</math>, Fraction <math>\theta/360</math>, Calculated area). Groups cut out each sector carefully, arrange the five cut-outs side by side on their desk in order from smallest to largest, and then fill in columns 1–3 of the table by measuring the angle (confirm with protractor) and radius of each sector. They also compute <math>\pi r^2 = \pi \times 7^2 \approx 154 \text{ cm}^2</math> together as a group before proceeding to column 4.</p> | <p>Distribute materials and say: 'Each group, cut out your five sectors carefully — scissors away from your body. Once cut, line them up on your desk from the smallest angle to the largest. What do you notice just by looking?' WAIT TIME: 15 seconds after groups line up sectors, then cold-call 2 groups: 'Group Kisumu, describe what you see. ... Group Mombasa, does your arrangement look the same?' Then direct: 'Now measure the angle of each cut-out with your protractor to confirm it, and measure the radius with your ruler. Record these in columns 1 and 2 of your table.' Circulate. At each group, ask: 'What is the full circle area for a circle of radius 7 cm? Show me how you worked it out.' WAIT TIME: 20 seconds before prompting if needed. If a group is stuck, say: 'What is the formula for the area of a full circle?' — not the answer. Ensure every group writes <math>\pi r^2 = \pi \times 49 \approx 154 \text{ cm}^2</math> in column 3 before moving on. After 8 minutes, pause the class and say: 'Hold up</p> | <p>Physical manipulation and data collection (Concrete–Representational–Abstract progression): Cutting and arranging physical sectors grounds the abstract formula in tangible objects. The five cut-outs on the desk become a concrete data set that learners can see, touch, and compare before any algebraic generalisation is made. Measuring to confirm pre-marked angles reinforces the meaning of central angle.</p> | <p>Circulate and check two specific observable indicators: (1) Every group can state and write <math>\pi r^2 = 154 \text{ cm}^2</math> correctly before moving to column 4 — if not, the formula for full circle area is not secure and must be addressed immediately. (2) When groups arrange sectors, check they place them in ascending order of angle (<math>30^\circ</math>, <math>60^\circ</math>, <math>90^\circ</math>, <math>120^\circ</math>, <math>180^\circ</math>) — this shows they understand angle determines size. If any group places sectors randomly, ask: 'Which sector has the smallest angle? Place that one first.'</p> |

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| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of a circle</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of a circle</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Groups complete columns 4 and 5 of their recording table: for each sector, they calculate the fraction <math>\theta/360</math> and then multiply by 154 cm<sup>2</sup> to find the sector area. They record results neatly in the table. A class data-sharing moment follows: the teacher builds a shared class table on the board as each group calls out their results. Learners look for the pattern across the five rows and write a sentence in their books: 'The area of a sector = ____.' Groups then apply the formula to two new problems: (A) A circular ugali cake of radius 14 cm is cut into a 45° slice for a guest — what is the area of the slice? (B) When a door in a Kisumu office opens 120°, it sweeps a sector of radius 80 cm on the floor — what area of floor does it sweep?</p> | <p>Say: 'Now complete columns 4 and 5. For column 4, divide your angle by 360. For column 5, multiply that fraction by 154. You have 5 minutes.' Circulate and, at each group, ask: 'For your 90° sector — what fraction of the full circle is it? ... So what is its area?' WAIT TIME: 30 seconds per question. Cold-call the answer from the group, not just the nearest learner. After 5 minutes, build the class table on the board by cold-calling each group for one row: 'Group Nairobi, give me the row for 60°.' Fill the table. Then ask the whole class: 'Look at the five rows. What pattern do you see between the angle and the area?' WAIT TIME: 30 seconds of silent individual thinking, then pair-share for 1 minute. Cold-call 3 pairs: 'Kendi and Otieno, what pattern did you find?' Guide toward the formula from Lesson 2: 'Area = <math>(\theta/360) \times \pi r^2</math>.' Write this on the board and say: 'Write this formula in your own words as a sentence in your exercise book.' Then present problems A and B: 'The mama mboga has a new problem. A customer wants the 45° ugali-cake slice — exactly how much ugali in cm<sup>2</sup> does she get? And a Kisumu office needs to know how much floor space the door sweeps. Use our formula.' Groups work for 5 minutes; cold-call one group per problem to present their working on the board.</p> | <p>Pattern recognition from a class data table (Data-driven generalisation): By pooling all five data points into a single shared table, learners see the linear relationship between <math>\theta</math> and area across the full range 30°–180°. Writing the formula as a sentence before using it algebraically ensures conceptual ownership rather than rote copying. Applying the formula to two new Kenyan contexts (ugali cake, office door) immediately tests transfer and connects to the driving question's real-life dimension.</p> | <p>Check two things during the apply phase: (1) Both problems must show the substitution step written out (e.g., Area = <math>(45/360) \times \pi \times 14^2</math>) — if a group writes only the final answer, ask: 'Show me every step.' (2) Listen for whether learners simplify 45/360 to 1/8 correctly. A common error is treating <math>\theta/360</math> as <math>\theta \times 360</math> — if this appears, draw the class's attention to it: 'What fraction of a pizza is one slice if you cut it into 8 equal pieces?' Use this as a whole-class re-teaching moment if more than 1 group makes this error.</p> |

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| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of a circle intuition</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Radians &amp; arc length</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Each group receives two sticky notes (or writes on the class DQB section of the board). On the first sticky note (GREEN), they write one thing they can now answer about the driving question: 'We can now calculate the area of any sector if we know <math>r</math> and <math>\theta</math>.' On the second sticky note (ORANGE), they write one new question or remaining puzzle that the lesson raised — for example: 'But what about the curved segment below a chord? That's not the same as a sector.' The teacher reads selected notes aloud and organises them on the DQB under columns: 'What we have figured out' and 'What we still need to figure out.'</p>                                                               | <p>Direct learners to the DQB. Say: 'We have been building our answer to the big question about Wakulima Market. On your green note, write one specific thing you can NOW calculate that you couldn't at the start of Lesson 1. Be precise — use the word formula or the word fraction.' WAIT TIME: 45 seconds of silent writing. Then: 'On your orange note, write one thing that our formula still cannot solve — think about the watermelon rind shape that is left over after you remove the triangular part of the slice.' WAIT TIME: 30 seconds. Cold-call 2 green notes and 2 orange notes. Post them on the DQB. Say: 'Notice that several groups are asking about segments — the region between a chord and an arc. That is exactly what we investigate in Lesson 5. We are building our answer piece by piece, just like the mama mboga builds her price by piece.'</p> | <p>Driving Question Board update (metacognitive progress tracking): The two-colour sticky note protocol makes learners' growing understanding visible and public. Distinguishing 'figured out' from 'still need to figure out' mirrors scientific practice — no single lesson answers everything. The teacher's act of reading and categorising notes models how scientists track the state of their knowledge and identify the next investigative step.</p> | <p>Read the green notes for specificity — acceptable notes reference the formula <math>\text{Area} = (\theta/360) \times \pi r^2</math> or a specific calculation. Vague notes like 'we learned about sectors' indicate surface engagement. If more than half the green notes are vague, ask the class: 'Can someone give me a green note that includes a number or a formula?' — prompt revision before posting. Orange notes that mention segments or chords confirm readiness for Lesson 5 and can be highlighted as the bridge.</p>                                                                   |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Arc length as fraction of circumference</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Arc length</a><br/> <i>Related topic: not an exact match for this lesson.</i></p>                                                                                                                                                                                                          | <p>Learners return to the annotated diagram of the Wakulima Market watermelon they began building in Lesson 1 (or draw a fresh circle of radius 7 cm if it is their first model). They now add three new annotations: (1) Label any two sectors on their diagram with their angle and calculated area (e.g., '90° sector → Area = 38.5 cm<sup>2</sup>'). (2) Write the general formula <math>\text{Area} = (\theta/360) \times \pi r^2</math> inside or beside the circle. (3) Draw an arrow from the formula to the phenomenon caption and write: 'This formula tells the mama mboga exactly how much watermelon (in cm<sup>2</sup>) any wedge contains — so every customer can be charged fairly by area, not by eye.' Groups share</p> | <p>Say: 'Open your model diagram from Lesson 1. We are going to update it with today's evidence. You have 4 minutes to add three things — I am putting them on the board.' Write on the board: '1. Label 2 sectors with angle AND area. 2. Write the formula <math>\text{Area} = (\theta/360) \times \pi r^2</math>. 3. Write one sentence connecting the formula to the mama mboga's problem.' Circulate and, for each group, ask: 'Read me your sentence connecting the formula to the market. Does it answer the driving question?' WAIT TIME: 20 seconds after each prompt. After 4 minutes, say: 'Swap your model with the group next to you. Use this sentence: Your model clearly</p>                                                                                                                                                                                      | <p>Cumulative annotated model revision (progressive model building): Returning to the same diagram across lessons externalises conceptual growth — learners can literally see what they have added each lesson. The peer feedback protocol using a structured sentence stem introduces scientific peer review norms and ensures models are evaluated against the driving question, not just aesthetics.</p>                                                  | <p>Collect or photograph 2–3 group models. Check: (1) Formula is written correctly — <math>\text{Area} = (\theta/360) \times \pi r^2</math>, not a variant like <math>\text{Area} = \theta/360 \times r</math>. (2) The two labelled sectors show correct numerical areas (e.g., 90° → 38.5 cm<sup>2</sup>, 60° → 25.67 cm<sup>2</sup>). (3) The connecting sentence explicitly mentions 'fair', 'area', or 'customer' — indicating the learner is connecting the mathematics to the phenomenon, not treating it as an abstract exercise. Use model photographs as evidence for portfolio assessment.</p> |

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| Source: Khan Academy (English - US curriculum)<br><a href="#">Search ARES for similar readings</a> | their updated model with the group next to them and give one piece of feedback using the sentence stem: 'Your model clearly shows ____, and it could be improved by ____.' | shows ____, and it could be improved by ____.' You have 2 minutes.' Cold-call 1 group to share the feedback they received. Close by saying: 'Our model is growing. Today we added the exact formula. In Lesson 5 we will add segments — the curved sliver of rind — and our model will be even more powerful.' |  |  |
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#### D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Did all five groups successfully derive  $\text{Area} = (\theta/360) \times \pi r^2$  from their data, or did some groups need the formula supplied? If more than 2 groups could not identify the pattern independently, consider whether the recording table needs a hint column (e.g., a pre-filled row for  $180^\circ$  showing  $180/360 = 1/2$ ) in future. (2) Were learners able to transfer the formula to the ugali cake and door-sweep problems? If transfer was weak, the apply phase needs a worked example before independent group practice. (3) Did the DQB orange notes mention segments? If not, learners may not yet be distinguishing sectors from segments — this distinction must be made explicit at the start of Lesson 5 before moving to the segment formula. (4) Were scissors managed safely? Note any group that needed additional safety reminders for the school's records. (5) Review the 2–3 collected model diagrams: are the phenomenon connections in the sentence annotations genuine, or are they formulaic? Genuine connections (mentioning fairness, price, or customer) indicate the phenomenon is driving the learning effectively.

#### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <b>What did I observe?</b>                   | Learners cut out five sectors of angles $30^\circ$ , $60^\circ$ , $90^\circ$ , $120^\circ$ , and $180^\circ$ from a printed circle of radius 7 cm, arranged them in order, measured their angles and radii, and recorded the full circle area ( $\pi r^2 \approx 154 \text{ cm}^2$ ) as the baseline.                                                                                                                                           |
| <b>What did I learn?</b>                     | The area of a sector is a fraction of the full circle area determined by the central angle: $\text{Area of sector} = (\theta/360) \times \pi r^2$ . A $90^\circ$ sector contains exactly one quarter of the full circle area; a $180^\circ$ sector contains exactly half. Changing only the angle — while keeping the radius constant — changes the area proportionally.                                                                        |
| <b>How does this explain the phenomenon?</b> | Learners applied the formula to two new Kenyan contexts (a $45^\circ$ ugali cake slice with $r = 14 \text{ cm}$ , and a door sweep of $120^\circ$ with $r = 80 \text{ cm}$ in Kisumu), updated their cumulative watermelon model with the formula and labelled sector areas, and added the connecting sentence: 'This formula tells the mama mboga exactly how much watermelon any wedge contains so every customer is charged fairly by area.' |

## LESSON 5 (40 minutes): Lesson 5 — From Wedge to Segment: Deriving and Applying the Area of a Circular Segment

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>Purpose</b>              | Learners derive the formula for the area of a circular segment by recognising it as the difference between a sector and a triangle, then apply this to determine the true area of the curved rind-bounded piece left when Mama Wanjiku's watermelon slice is cut straight across the chord.                                                                                                                                                                                                                                                      |
| <b>Knowledge</b>            | <ul style="list-style-type: none"> <li>– A circular segment is the region between a chord and the arc it cuts off.</li> <li>– The area of a segment = Area of sector – Area of triangle, i.e. <math>A = (\theta/360) \times \pi r^2 - \frac{1}{2}r^2 \sin \theta</math>.</li> <li>– The triangle inside a sector formed by two radii and a chord has area = <math>\frac{1}{2}r^2 \sin \theta</math>, where <math>\theta</math> is the central angle in degrees.</li> </ul>                                                                       |
| <b>Skills</b>               | <ul style="list-style-type: none"> <li>– Calculate the area of the isosceles triangle formed by two radii and a chord using <math>A = \frac{1}{2}r^2 \sin \theta</math>.</li> <li>– Derive and apply the segment area formula <math>A = (\theta/360) \times \pi r^2 - \frac{1}{2}r^2 \sin \theta</math> to solve real-life problems involving curved shapes.</li> </ul>                                                                                                                                                                          |
| <b>Attitudes</b>            | <ul style="list-style-type: none"> <li>– Appreciate that breaking a complex shape into familiar parts (sector and triangle) is a powerful mathematical problem-solving strategy.</li> <li>– Demonstrate curiosity and persistence when moving from the known sector formula to the new segment formula.</li> </ul>                                                                                                                                                                                                                               |
| <b>Key Inquiry Question</b> | Where do we use the area of part of a circle — specifically a segment — in real life, and how do we calculate it exactly?                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Purpose in Storyline</b> | Lessons 2–4 established that the area of a sector = $(\theta/360) \times \pi r^2$ . This lesson advances the storyline by asking: what if Mama Wanjiku cuts straight across the watermelon, creating a flat edge rather than a pointed wedge? That curved rind-bounded piece is a segment — a new shape the learners cannot yet calculate. By discovering that a segment = sector – triangle, learners unlock the second part-of-a-circle formula and move closer to fully solving any fair-sharing or design problem involving circular shapes. |
| <b>Safety Notes</b>         | No sharp instruments required. If physical circular cut-outs are used, scissors should be handled with care and passed handle-first. No other specific hazards.                                                                                                                                                                                                                                                                                                                                                                                  |

### B. LESSON OVERVIEW

This lesson is the fifth in the evidence-gathering sequence and marks a conceptual turning point in the sub-strand storyline. Having derived the sector area formula in Lesson 2 and practised it in Lessons 3–4, learners now encounter a genuinely new shape: the circular segment — the region trapped between a chord and its arc. The lesson is anchored in the Wakulima Market phenomenon: when Mama Wanjiku decides to cut straight across a watermelon wedge (slicing off the pointed tip along a flat chord), she creates a curved, rind-bounded piece whose area her customers cannot estimate using any formula learned so far. This creates the need for a new tool.

The lesson is structured around a guided discovery: learners first predict whether the segment is larger or smaller than the triangle that fits inside the same sector, then observe a visual decomposition (sector = triangle + segment, therefore segment = sector – triangle). Through the Explain phase, learners derive the segment formula step by step — Area of sector =  $(\theta/360) \times \pi r^2$ , Area of triangle =  $\frac{1}{2}r^2 \sin \theta$ , therefore Area of segment =  $(\theta/360) \times \pi r^2 - \frac{1}{2}r^2 \sin \theta$  — and apply it to two Wakulima Market problems. The  $\frac{1}{2}r^2 \sin \theta$  triangle formula is introduced here, building on the two-sides-and-included-angle area formula from Sub-Strand 2.4 Area of Polygons.

By the end of the lesson learners update their Driving Question Board with the segment formula and revise their cumulative circle-part model to include both sectors and segments, clearly annotating the relationship between the two. Teacher moves are designed for a 40-minute period with deliberate wait times, targeted cold-calling, and

a structured pair-share to ensure every learner constructs — rather than merely copies — the new formula.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                  | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Area of Sectors and Segments Principles - Basic</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: CK-12<br> <a href="#">Search ARES for similar videos</a><br><br> <b>HTML:</b><br><a href="#">Area of Sectors and Segments (Flexbook)</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | Learners are shown a large circle diagram on the board (or a printed hand-out) in which a chord has been drawn across a sector of radius 14 cm and central angle $60^\circ$ . The sector is shaded; the triangle inside it is outlined in a different colour. Learners individually write answers to two questions in their exercise books: (1) 'Which is larger — the shaded sector or the triangle inside it? By how much, roughly?' and (2) 'If I remove the triangle from the sector, what shape is left, and how would you describe its boundary?' After 2 minutes of silent individual writing, learners share predictions with a seat partner for 1 minute. | Display the diagram and read aloud: 'Look carefully at this circle. A chord has been drawn inside a sector. In your books, answer these two questions — work alone and in silence for two minutes. I want YOUR thinking, not your neighbour's.' [Allow 2 minutes of silent thinking — do not circulate or give hints during this time.] After 2 minutes: 'Now turn to your partner and share what you wrote. You have one minute — go.' [After 1 minute, cold-call 3 pairs — choose one stronger and two mid-range learners.] Ask pair 1: 'What did you both agree the leftover shape looks like?' [Wait 10 seconds.] Ask pair 2: 'Did you think the triangle was bigger or smaller than the sector? What was your reasoning?' [Wait 12 seconds.] Ask pair 3: 'Can you describe the boundary of that leftover shape in your own words?' [Wait 10 seconds.] Do NOT confirm or correct yet. Say: 'Hold those ideas — we are about to find out who is right.' | Predict–Observe–Explain (POE) launch: By asking learners to commit a written prediction before any instruction, this strategy activates prior knowledge of sectors and triangles, surfaces misconceptions about curved boundaries, and creates genuine cognitive need to observe the decomposition. The prediction becomes a personal stake in the upcoming evidence. | Teacher scans written responses during the silent phase, noting (a) whether learners correctly identify the sector as larger than the triangle, and (b) whether they can describe the segment boundary as 'an arc and a straight line (chord)'. Learners who describe the leftover shape as 'another triangle' or 'a rectangle' signal a gap in understanding curved boundaries — flag these learners for targeted questioning in the Observe phase. |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Area of Sectors and Segments Principles - Basic</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: CK-12<br> <a href="#">Search ARES</a>                                                                                                                                                                                                                                                                                                                                        | Learners receive (or draw) a circle of radius 14 cm with a $60^\circ$ sector clearly marked. They physically cut out (or shade-and-label on paper) three regions: the full sector, the isosceles triangle formed by the two radii and the chord, and the remaining curved piece (the segment). Working in groups of three, learners measure the chord                                                                                                                                                                                                                                                                                                              | Distribute cut-out circles (or direct learners to draw a 14 cm radius circle using compasses). Say: 'In your group of three, cut out — or shade on paper — the sector with angle $60^\circ$ . Then draw the chord connecting the two ends of the arc. Now you have three regions. Label them: call the big curved slice SECTOR, call the pointy triangle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Decomposition and Reassembly (Breaking Complex into Known Parts): Learners physically or visually separate the segment from the sector, making the abstract relationship Segment = Sector – Triangle concrete and manipulable. This strategy mirrors the NGSS Science and Engineering Practice of 'Analysing                                                          | Teacher checks that every group has correctly written 'Segment = Sector – Triangle' before moving on. Key observable indicators: (1) The chord is drawn as a straight line connecting the two arc endpoints — not a curve. (2) The segment is identified as the region on the arc side of the chord, not the triangle side. (3) The sector                                                                                                           |

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| <p><a href="#">for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of Sectors and Segments Practice</a><br/> Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                         | <p>length using a ruler, label each region, and record: 'Sector = Triangle + Segment, therefore Segment = Sector – Triangle.' They then calculate the sector area using the formula they already know — <math>(60/360) \times \pi \times 14^2</math> — and record the numerical result. They also note the shape of the triangle (two equal sides of 14 cm, included angle <math>60^\circ</math>, making it equilateral) and estimate the triangle's area by counting squares on graph paper or by recalling the triangle area formula from earlier work.</p>                                                                                                                               | <p>TRIANGLE, and call the curved bit left between the chord and the arc SEGMENT.' [Circulate for 3 minutes, checking labelling. If a group is stuck: 'Show me where the chord is. Good. Everything on the arc side of that chord — what shape is that?'] After groups label correctly, say: 'Now write this relationship in your books: Sector = Triangle + Segment. Rearrange it — what does Segment equal?' [Wait 12 seconds. Cold-call one group who has not yet spoken today.] 'Correct — Segment = Sector minus Triangle. Now calculate the sector area using the formula we proved in Lesson 2. Write the substitution step, not just the answer.' [Allow 3 minutes. Circulate and check substitution is written as <math>(60/360) \times \pi \times 14^2</math>.] Say: 'Look at your triangle. Its two equal sides are both radii — 14 cm each — and the angle between them is <math>60^\circ</math>. What kind of triangle is that?' [Wait 10 seconds. Cold-call a learner who predicted incorrectly in the Predict phase.] 'Yes — equilateral. We will use that in the next phase to find its area exactly.'</p> | <p>and Interpreting Data' — learners extract a structural relationship from a physical object rather than receiving it as a rule.</p>                                                                                                                                                                                                                                                                                                                                                | <p>area is calculated with a written substitution step showing <math>(60/360) \times \pi \times 14^2</math>. Groups that shade the wrong region (the major segment instead of the minor) should be redirected immediately with the question: 'Which piece is smaller — the one with the arc close to the chord, or the big curved piece? We are working with the smaller one.'</p>                                                                                                                                                                                                                        |
| <p><b>Explain Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Area of Sectors and Segments Principles - Basic</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Area of Sectors and Segments (Flexbook)</a><br/> Source: CK-12</p> | <p>The class derives the triangle area formula <math>\frac{1}{2}r^2\sin \theta</math> together, guided by the teacher. Learners substitute values (<math>r = 14</math>, <math>\theta = 60^\circ</math>) to find the triangle area, then subtract from the sector area to find the segment area. They record the general formula: Area of segment = <math>(\theta/360) \times \pi r^2 - \frac{1}{2}r^2\sin \theta</math>. Learners then individually solve two Wakulima Market problems: (A) Mama Wanjiku cuts a <math>90^\circ</math> sector from her 14 cm radius watermelon and then slices off the pointed tip along a chord — find the segment area; (B) A market stall designer in</p> | <p>Write on the board: 'We know: Area of triangle = <math>\frac{1}{2} \times \text{base} \times \text{height}</math>. For our isosceles triangle with two sides of length <math>r</math> and included angle <math>\theta</math>, there is a trigonometry shortcut.' Ask: 'From Sub-Strand 2.4, if I know two sides and the included angle of a triangle, what formula gives me the area?' [Wait 15 seconds. Cold-call a learner from the Trigonometry unit.] If no response: 'The formula is <math>\frac{1}{2} \times a \times b \times \sin C</math>. Here, <math>a = b = r</math> and <math>C = \theta</math>. So area of our triangle = <math>\frac{1}{2} \times r \times r \times \sin \theta = \frac{1}{2}r^2\sin \theta</math>.' Write this step by step. Say: 'Now</p>                                                                                                                                                                                                                                                                                                                                             | <p>Worked Example → Generalise → Apply (Gradual Release): The teacher models one full numerical example with the class, then immediately releases learners to apply the generalised formula to two new contexts. Connecting <math>\frac{1}{2}r^2\sin \theta</math> to the already-known two-sides-and-included-angle triangle formula (Sub-Strand 2.4 Area of Polygons) makes the new step a logical extension rather than an isolated rule, reinforcing cross-strand coherence.</p> | <p>Teacher looks for three specific things in written working: (1) The formula is stated before substitution — not just a numerical calculation. (2) <math>\sin 90^\circ = 1</math> is correctly applied in Problem A (many learners mistakenly write <math>\sin 90^\circ</math> as 0.9 or omit it). (3) Units are written as <math>\text{cm}^2</math>. For Problem B, a common error is subtracting the triangle area before computing it — teacher watches for premature subtraction. Any learner who gets a negative segment area has subtracted in the wrong order and needs immediate one-to-one</p> |

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| <p><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                               | <p>Kenyatta Market, Nairobi, wants to tile a curved decorative panel that is a segment of a circle with radius 20 cm and central angle <math>120^\circ</math> — find the area of the panel. Learners write full working, including formula statement, substitution, and answer with units.</p>                                                                                                                                                                                                       | <p>substitute <math>r = 14</math>, <math>\theta = 60^\circ</math> and calculate <math>\frac{1}{2} \times 14^2 \times \sin 60^\circ</math>. Use your calculator. Write the substitution first.' [Allow 90 seconds. Walk to three different tables to check calculator use and substitution.] Cold-call one learner for the numerical answer. Confirm: <math>\frac{1}{2} \times 196 \times 0.8660 \approx 84.87 \text{ cm}^2</math>. Then say: 'Now subtract your triangle area from your sector area to get the segment area. Do this in your books.' [Wait 2 minutes.] 'So our general formula is...' [write slowly, pointing to each term]: 'Area of segment = <math>(\theta/360) \times \pi r^2 - \frac{1}{2}r^2 \sin \theta</math>.' Say: 'Copy this formula with a box around it. It is the second major formula of this sub-strand.' Then: 'You have 6 minutes — solve both Wakulima Market and Kenyatta Market problems in your books. Show every step. Problem A: <math>90^\circ</math>, <math>r = 14 \text{ cm}</math>. Problem B: <math>120^\circ</math>, <math>r = 20 \text{ cm}</math>. Go.' [Circulate; after 6 minutes cold-call two different learners to present their working for A and B respectively at the board. For each presenter: 'Talk us through your substitution step by step.' Correct errors publicly and positively.]</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>redirection.</p>                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p><b>Driving Question Board (DQB) Creation</b><br/>  <b>VIDEO:</b><br/> <a href="#">Area of Sectors and Segments Principles - Basic</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/> <a href="#">Search ARES for similar videos</a></p> | <p>Learners return to the class Driving Question Board. Each learner writes one sticky note (or contributes to a shared poster) responding to the prompt: 'What new tool did we add today for calculating a part of a circle, and when would Mama Wanjiku need it instead of the sector formula?' Learners post their notes under the column 'What We Now Know'. The class then reads three posted notes aloud and discusses whether the DQB driving question — about calculating the exact area</p> | <p>Direct learners to the DQB: 'We started this unit with Mama Wanjiku's watermelon at Wakulima Market. Our driving question asked about wedge-shaped AND segment-shaped pieces. Until today, we could only handle wedge shapes — sectors. What changed today?' [Wait 12 seconds. Take 3 raised hands — not cold-call here, as this is a public knowledge-building moment.] After responses: 'Write on your sticky note: one formula and one real-life situation where you would use a</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Public Knowledge Aggregation (Driving Question Board Update): The DQB functions as a living class document that maps growing understanding against the original phenomenon. Posting sticky notes externalises individual thinking and makes patterns of understanding — and gaps — visible to the whole class and teacher simultaneously. Revisiting the driving question at this stage reinforces that each lesson is a step in an ongoing investigation, not an isolated topic.</p> | <p>Teacher reads all posted sticky notes during or immediately after the DQB phase. Indicators of strong understanding: learner correctly identifies that a segment needs a chord (flat cut), whereas a sector needs two radii meeting at a point. Indicators of confusion: learner conflates segment and sector, or cannot state when each is used. These learners are identified for targeted follow-up in the next lesson's Predict phase.</p> |

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| <p> <b>EXERCISE:</b><br/> <a href="#">Area of Sectors and Segments Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>of any wedge-shaped or segment-shaped piece — can now be more fully answered. A class scribe adds the segment formula to the 'Our Growing Formula Bank' section of the DQB.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>segment rather than a sector. You have 90 seconds — write now.' [After 90 seconds, direct learners to post notes.] Read three notes aloud — choose one strong, one partial, one that contains a misconception. For the misconception note (e.g. 'use segment whenever you have a circle'), say: 'This learner is asking a great implied question — when exactly do you have a segment? Let's clarify: a segment appears when...' [Prompt class.] After: 'Scribe, please add today's formula to the Formula Bank section.' [Dictate clearly: Area of segment = <math>(\theta/360) \times \pi r^2 - \frac{1}{2}r^2 \sin \theta</math>.]</p>                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Model Building Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Area of Sectors and Segments Principles - Basic</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Chords in Circles: Arc and Chord (PLIX)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners open their cumulative 'Circle Parts Model' — a diagram they have been building across Lessons 1–4 showing a circle divided into sectors with annotations of the sector formula. Today they add a new layer: they draw a chord across one sector, shade the resulting segment, and annotate it with the segment formula. They also add a 'decision tree' annotation beside their model: 'If the shape has two radii and an arc → use SECTOR formula. If the shape has one chord and an arc → use SEGMENT formula.' Learners write a one-sentence 'model update statement' beneath their diagram explaining what new understanding the segment formula adds to their ability to answer the driving question.</p> | <p>Say: 'Take out your Circle Parts Model from your portfolio. You have been building this since Lesson 1. Today we add the segment.' Give explicit instructions: 'Step 1 — draw a chord across one of your existing sectors. Step 2 — shade the segment (the region between the chord and the arc). Step 3 — write the segment formula as an annotation with an arrow pointing to the shaded region. Step 4 — add the decision tree I am about to write on the board.' [Write on board: TWO RADII + ARC → SECTOR → <math>A = (\theta/360)\pi r^2</math>. ONE CHORD + ARC → SEGMENT → <math>A = (\theta/360)\pi r^2 - \frac{1}{2}r^2 \sin \theta</math>.] 'You have 4 minutes for steps 1–4. Begin.' [Circulate; after 4 minutes cold-call one learner: 'Read me your model update statement.' Then ask a second: 'In your model, point to the segment and tell me the two parts you subtract to find its area.' Wait 10 seconds each time.] Close with: 'Today Mama Wanjiku's customers can now ask</p> | <p>Cumulative Model Revision (Progressive Diagram Annotation): The circle model serves as a running scientific model that is revised as understanding deepens — directly mirroring the NGSS practice of 'Developing and Using Models'. The decision tree annotation is a metacognitive tool that helps learners self-monitor which formula to select in novel problems, building flexible procedural knowledge rather than rote application.</p> | <p>Teacher checks that (1) the chord is drawn as a straight line across an existing sector (not a new freehand curve), (2) the segment is shaded on the arc side of the chord (not the triangle side), (3) both the formula annotation and the decision tree are present and correctly written. The one-sentence model update statement is the richest formative indicator: a complete sentence such as 'The segment formula lets me find the area of any curved piece of a circle that is bounded by a chord and an arc, not just a wedge' shows full integration; a statement such as 'I learned the segment formula' shows surface engagement only.</p> |

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|  |  | about ANY curved piece of her watermelon — wedge or flat-cut slice — and we have the exact formula to answer them. Next lesson we will use both formulas together to solve a full fair-sharing design problem.' |  |
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### D. TEACHER REFLECTION

1. During the Explain phase, did ALL learners correctly apply  $\sin 90^\circ = 1$  in Problem A, or did a significant number treat it as a decimal approximation? If the latter, how will I address the exact trigonometric values ( $\sin 30^\circ$ ,  $\sin 60^\circ$ ,  $\sin 90^\circ$ ) at the start of Lesson 6?
2. When learners updated the Driving Question Board, how many sticky notes demonstrated a clear distinction between a sector and a segment (two radii vs. one chord)? Were there learners who still conflated the two shapes, and how will I re-expose them to that boundary condition?
3. Did the connection to Sub-Strand 2.4 ( $\frac{1}{2}ab \sin C$ ) feel natural to learners, or did it seem like a new and unrelated formula? What does this tell me about the extent to which learners are building an integrated, cross-strand mathematical toolkit rather than isolated rules?
4. Was the 40-minute period sufficient for learners to complete both Wakulima Market problems with full written working, or did time pressure cause learners to skip the formula-statement step? If many skipped this step, what classroom norm do I need to reinforce?
5. Looking at the cumulative Circle Parts Models learners updated today, are the decision trees accurate and do they reflect genuine understanding of when to choose each formula? Which learners' models suggest they are ready for the combined-problem challenge planned for Lesson 6, and which learners need a consolidation activity before moving on?

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>What did I observe?</b>                   | Learners observed that when a chord is drawn across a sector of Mama Wanjiku's 14 cm radius watermelon, the sector is divided into two distinct regions — an isosceles triangle (formed by the two radii and the chord) and a curved segment (the region between the chord and the arc). They saw physically and numerically that $\text{Segment} = \text{Sector} - \text{Triangle}$ , and verified this with the specific case of a $60^\circ$ sector.                                                                                                    |
| <b>What did I learn?</b>                     | The area of a circular segment is found by subtracting the area of the isosceles triangle from the area of the sector that contains it: $\text{Area of segment} = (\theta/360) \times \pi r^2 - \frac{1}{2}r^2 \sin \theta$ . The triangle's area is calculated using the two-sides-and-included-angle formula $\frac{1}{2}r^2 \sin \theta$ , which connects to the triangle area work in Sub-Strand 2.4 Area of Polygons. A segment is bounded by one chord and one arc; a sector is bounded by two radii and one arc — these require different formulas. |
| <b>How does this explain the phenomenon?</b> | The phenomenon involves two types of curved pieces: the pointed wedge (sector) and the flat-cut curved rind-piece (segment). Now that learners have both the sector formula (Lessons 2–4) and the segment formula (Lesson 5), they can calculate the exact area of ANY piece Mama Wanjiku cuts from her watermelon — whether she cuts along two radii (a sector) or along a chord (a segment). This means a customer at Wakulima Market can now verify mathematically, not just visually, whether their flat-cut curved slice is a fair share.             |

## LESSON 6 (80 minutes): Applying the Sector Area Formula: From Watermelon Wedges to Irrigation Arcs

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Purpose</b>              | To formalise the sector area formula and apply it fluently to graduated real-life Kenyan contexts — from a watermelon slice at Wakulima Market to a school gate swing to a centre-pivot irrigation arc — so that learners can calculate the exact area of any sector and justify their reasoning to peers.                                                                                                                                                                                        |
| <b>Knowledge</b>            | Learners will know: (a) the sector area formula $A = (\theta/360) \times \pi r^2$ for a central angle $\theta$ in degrees; (b) that sector area scales proportionally with both the angle $\theta$ and the square of the radius $r$ ; (c) how to identify radius and central angle from real-life sector-shaped objects in Kenya.                                                                                                                                                                 |
| <b>Skills</b>               | Learners will be able to: (a) recall and state the sector area formula derived in Lesson 2; (b) substitute values accurately into $A = (\theta/360) \times \pi r^2$ and compute area in $\text{cm}^2$ , $\text{m}^2$ , and hectares; (c) present and evaluate worked solutions in pairs and as a class; (d) update the Driving Question Board with the formalised formula and a new arising question.                                                                                             |
| <b>Attitudes</b>            | Learners will appreciate: (a) the power of a single formula to resolve the 'unfair watermelon slice' dispute at Wakulima Market; (b) the role of mathematics in fair-sharing and real-world design problems; (c) the value of collaboration and peer evaluation in consolidating mathematical understanding.                                                                                                                                                                                      |
| <b>Key Inquiry Question</b> | Where do we use area of part of a circle in real life?                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Purpose in Storyline</b> | In Lessons 2–4 learners derived, confirmed and practised the sector formula, and Lesson 5 added segments. In Lesson 6 the teacher consolidates the sector formula, anchors it to three graduated Kenyan contexts (watermelon slice, school gate arc, irrigation pivot), and learners practise applying it in pairs before updating the DQB. This lesson moves the class from evidence-gathering to explanation, directly answering the driving question's demand for an exact calculation method. |
| <b>Safety Notes</b>         | No significant hazards. If physical circular cut-outs or compasses are used for illustration, remind learners to handle compass points with care. Ensure adequate desk space when pairs spread out workbooks for peer evaluation.                                                                                                                                                                                                                                                                 |

### B. LESSON OVERVIEW

Lesson 6 of 8 in the evidence-gathering sequence. The teacher opens by projecting the Wakulima Market watermelon phenomenon and asking pairs to recall the sector formula derived in Lesson 2. After a brief Predict phase, the teacher revisits  $A = (\theta/360) \times \pi r^2$  through a whole-class recap of its derivation anchored to the phenomenon. Learners then work through three graduated Kenyan context examples — (1) a  $120^\circ$  watermelon slice of radius 15 cm, (2) a school gate of radius 1.2 m swinging through  $75^\circ$ , (3) a centre-pivot irrigation arc of radius 80 m through  $270^\circ$  — first as a guided worked example, then in pairs, then independently. Pairs present solutions on mini-whiteboards for class evaluation. The lesson closes with DQB updating: learners add the formalised formula and post one new question that has arisen. Teacher reflection data are gathered via an exit card. Connects directly to KICD SLO (a): determine the area of a sector of a circle in different situations, and SLO (c): apply the area of part of a circle in real-life situations.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>Learners open their exercise books to their Lesson 2 derivation of the sector area formula. Working individually for 2 minutes, each learner writes a prediction: 'Using the formula from Lesson 2, what area (in cm<sup>2</sup>) would a watermelon sector of radius 15 cm and angle 120° have?' They record their prediction and reasoning in one sentence. After individual thinking, pairs compare predictions and note whether they agree or differ, circling the part of their formula they are least confident about. Predictions remain visible on the desk throughout the lesson as a reference point.</p> | <p>Display the Wakulima Market image on the board (or sketch a large circle with two radii drawn at 120°, labelled 'watermelon, <math>r = 15</math> cm, <math>\theta = 120^\circ</math>'). Say verbatim: 'In Lesson 2 we derived a formula for sector area. Before I show you anything new, I want you to trust your own thinking. Individually — not with your partner yet — use that formula and predict the area of this 120° slice. Write your answer and one sentence explaining why.' Allow 2 minutes of silent individual work. Then say: 'Now turn to your partner. Compare your answers. If they differ, identify exactly which part of the formula caused the difference.' Allow 2 minutes of pair talk. Cold-call 3 pairs (choose one pair whose prediction is correct, one slightly off, one using a different but plausible approach). For each, ask: 'Walk us through your formula — where does each part come from?' WAIT TIME: 12 seconds after each cold-call before accepting response. Record the three predictions on the board in a column headed 'Our Predictions' without evaluating them yet.</p> | <p>Strategy: Predict-Before-Instruction (PBI) anchored to the phenomenon. By committing a specific numerical prediction to paper before instruction, learners activate their existing mental models from Lessons 1–5. Displaying competing predictions on the board creates productive cognitive dissonance — learners are motivated to find out which prediction the formalised formula will confirm, directly tying the formula derivation to a felt need.</p>                                                    | <p>Observable indicator: Every learner has a written numerical prediction and a reasoning sentence before pair talk begins. Teacher circulates during individual work (2 min) and notes: (a) which learners use a proportional structure (<math>\theta/360 \times</math> something) — these have the key conceptual anchor; (b) which learners attempt an additive or unrelated approach — these need targeted support during the Explain phase. Teacher uses this scan to select the three pairs for cold-call to represent the range of thinking in the room.</p> |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of a sector</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a></p>                   | <p>The teacher projects or draws a full circle of radius <math>r</math> and shades a sector of angle <math>\theta</math>. Learners observe as the teacher builds the derivation step-by-step on the board, with learners completing a structured note-frame in their books simultaneously. The note-frame has five gaps: (1) Area of full circle = ____; (2) A full circle has a central angle of ____°; (3) A sector with angle <math>\theta</math> is ____ of the full circle; (4) Therefore, area of sector = ____ <math>\times</math> area of full circle; (5) Simplified formula: <math>A =</math> ____.</p>      | <p>Say: 'Let's revisit how the Lesson 2 formula is built from first principles.' Draw a full circle, label radius <math>r</math>. Step 1 — write 'Full circle area = <math>\pi r^2</math>' and ask: 'Why <math>\pi r^2</math>? Shout it out.' (Accept responses, reinforce.) Step 2 — draw a sector, label <math>\theta</math>. Ask: 'If the full circle is 360°, what FRACTION of the circle is this sector?' WAIT TIME: 10 seconds. Cold-call 1 learner. Step 3 — write 'Sector area = <math>(\theta/360) \times \pi r^2</math>' on the board in a box. Say: 'Write this formula in your note-frame and say it out loud to your</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Strategy: Structured Note-Frame Derivation with Prediction Revisit. The note-frame ensures learners are not passive — they construct the formula in writing at the same time as the teacher. The immediate return to the prediction column closes the predict–observe loop: learners experience the formula as the resolution of their earlier uncertainty rather than as an imposed rule. The three-sector ranking task uses relational reasoning (proportionality) before computation, building conceptual</p> | <p>Observable indicator: All five gaps in the note-frame are completed correctly. Teacher does a 30-second visual scan of note-frames before moving on — specifically checking gap (3) ('<math>\theta/360</math> of the full circle') as this is the proportionality concept most likely to be incomplete. If more than 4 learners have gap (3) blank or incorrect, teacher pauses and re-teaches the fraction-of-a-full-turn idea using: 'If Amina gets 1 slice out of 3 equal slices, what fraction does she get? If her slice angle is 120° out of</p>           |

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| <p><a href="#">for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>After the derivation, learners observe three physical demonstrations of scale: teacher sketches a 30° sector, a 120° sector, and a 270° sector of the same radius on the board and asks learners to use the formula to rank them by area before computing — reconnecting to the watermelon vendor's problem of unequal-looking slices.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>partner twice.' (30 seconds.) Step 4 — return to the prediction column on the board. Say: 'Look at our three predictions from earlier. Using this formula, which prediction was correct? Put a tick next to the correct one in your own notes.' WAIT TIME: 12 seconds. Cold-call 2 learners to explain the tick. Step 5 — sketch the three sectors (30°, 120°, 270°, same r) and say: 'Before you calculate anything, rank these three sectors by area — smallest to largest. Write your ranking.' WAIT TIME: 10 seconds. Then ask 2 learners to justify their ranking. Confirm: 'Area grows with angle — this is exactly why the mama mboga's customers with 30° slices were complaining!'</p>                                                                                                                                                                                                                                                                                                                                                                                                | <p>understanding ahead of procedural practice.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>360°, what fraction is that?' before proceeding.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Area of a circle</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of a circle</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners work through three graduated examples in Kenyan contexts, progressing from guided (teacher-led) to paired to individual. EXAMPLE 1 (Guided, whole class — 8 min): Watermelon slice. A circular watermelon at Wakulima Market has radius 15 cm. A vendor cuts a sector of angle 120°. Calculate the exact area of the slice in cm<sup>2</sup>, giving the answer in terms of <math>\pi</math> and as a decimal (use <math>\pi = 3.142</math>). Learners follow the teacher's worked solution on the board, completing each calculation step in their books simultaneously. EXAMPLE 2 (Paired — 10 min): School gate. A metal school gate rotates about a hinge post. The gate arm is 1.2 m long. When fully opened, it sweeps through an angle of 75°. Calculate the area of ground swept by the gate in m<sup>2</sup>, correct to 3 significant figures. Pairs work side-by-side; one partner writes the</p> | <p>EXAMPLE 1: Write the problem on the board. Say: 'This is our mama mboga's exact slice — 120°, radius 15 cm. We are going to solve it together step by step. I write, you write.' Narrate each step aloud: 'Step 1 — identify what we know: <math>\theta = 120^\circ</math>, <math>r = 15</math> cm. Step 2 — write the formula: <math>A = (\theta/360) \times \pi r^2</math>. Step 3 — substitute: <math>A = (120/360) \times \pi \times 15^2</math>. Step 4 — simplify the fraction: <math>120/360 = 1/3</math>. Step 5 — work out <math>15^2 = 225</math>. Step 6 — <math>A = (1/3) \times \pi \times 225 = 75\pi</math> cm<sup>2</sup>. Step 7 — decimal: <math>75 \times 3.142 = 235.65</math> cm<sup>2</sup>.' Pause after each step for learners to write. After Step 4 ask: 'Why did 120/360 simplify to 1/3? What does that tell us about this slice?' WAIT TIME: 10 seconds. Cold-call 1 learner. After completing: 'So the customer's slice is exactly 235.65 cm<sup>2</sup>. If the vendor claims all slices are equal but another customer's slice is 180° — how large is that</p> | <p>Strategy: Gradual Release of Responsibility (I Do → We Do → You Do) within a Kenyan Context Progression. Each example escalates in two dimensions simultaneously: (1) computational demand (integer angle that simplifies → decimal angle → reflex angle with unit conversion), and (2) physical scale (kitchen-table watermelon → school compound gate → commercial farm). This progression re-anchors each calculation to the driving question's real-life scope, showing learners that one formula handles wedge-shaped problems from centimetres to metres to hectares. Pair presentation and class evaluation using explicit criteria (formula setup, arithmetic) make peer assessment a sensemaking activity rather than a social exercise.</p> | <p>Observable indicators: (1) During Example 1, all learners have seven written steps in their books — teacher scans before moving to Example 2. (2) During Example 2 circulation, teacher specifically checks whether the pair with the most errors in Lessons 1–5 (identified from prior lesson records) can write <math>A = (75/360) \times \pi \times 1.2^2</math> correctly. (3) After Example 3 peer marking, teacher tallies show-of-hands data: if more than 6 learners got part (a) wrong, teacher re-does the 270° substitution (common error: subtracting 270 from 360 first). If more than 8 learners got the unit conversion wrong, teacher spends 2 additional minutes on the hectare conversion with a second Naivasha farm example before closing.</p> |

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|                                     | <p>formula substitution, the other writes the numerical evaluation, then they swap roles to check. One pair is selected to present on a mini-whiteboard. Class evaluates using two criteria on the board: (i) correct formula substitution, (ii) correct arithmetic. EXAMPLE 3 (Independent — 12 min): Centre-pivot irrigation. A centre-pivot irrigation system near Naivasha has an arm of radius 80 m. The system currently operates over a sector of <math>270^\circ</math>. Calculate: (a) the area irrigated in <math>\text{m}^2</math>; (b) the area in hectares (1 hectare = <math>10,000 \text{ m}^2</math>). Learners work independently and silently. After 10 minutes, pairs peer-mark using the answer written on the board. Learners who finish early are challenged: 'The farmer wants to expand the system to cover the full circle. How many additional <math>\text{m}^2</math> would be irrigated?'</p> | <p>slice? Do the maths in your head.' WAIT TIME: 12 seconds. Cold-call 1 learner. EXAMPLE 2: Write the gate problem. Say: 'With your partner — one of you writes the formula and substitution, the other evaluates the arithmetic. Then swap and check each other. You have 8 minutes.' Circulate and target pairs where substitution errors appear (common error: using diameter instead of radius). Ask targeted prompt: 'The formula uses <math>r</math> — what is <math>r</math> here? Is 1.2 m the radius or the diameter?' After 8 minutes, invite one pair to present on a mini-whiteboard. Say to the class: 'Evaluate this solution on two criteria — correct formula setup, and correct arithmetic. Give a thumbs up for each criterion you agree with.' Address any arithmetic errors whole-class. EXAMPLE 3: Say: 'This one is yours — independent, silent. Read carefully — there are two parts, and part (b) requires a unit conversion.' Circulate silently. After 10 minutes say: 'Exchange books with your partner. I'll write the answers on the board — mark part (a) and part (b) separately.' Write: '<math>A = (270/360) \times \pi \times 80^2 = (3/4) \times 3.142 \times 6400 = 15,079.6 \text{ m}^2</math>; in hectares = 1.508 ha.' After peer marking ask: 'Raise your hand if you got both parts correct.' Then: 'Raise your hand if you got part (a) correct but made a unit conversion error in part (b).' Address unit conversion error directly: 'Dividing by 10,000 — is the answer going to be bigger or smaller than your <math>\text{m}^2</math> answer? Why?'</p> |                                                                                                     |                                                                                                    |
| <b>Driving Question Board (DQB)</b> | Learners return to the class Driving Question Board displayed at the front. Individually, each                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Say: 'For the last four minutes, I want two sticky notes from each of you — no sharing yet, individual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Strategy: DQB Incremental Update with Metacognitive Framing. The DQB is a public, cumulative record | Observable indicators: (1) Formula cards that use the watermelon context in their explanation (not |

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| <p><b>Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of a circle intuition</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Arc length</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>learner writes two sticky-note responses: (1) a FORMULA CARD — they write the sector area formula <math>A = (\theta/360) \times \pi r^2</math> in their own words and add a one-line explanation of what each symbol means in the context of the watermelon phenomenon (e.g., '<math>\theta</math> is the angle the vendor cuts, <math>r</math> is how far the rind is from the centre'); (2) a NEW QUESTION CARD — they write one new question that arose during today's lesson that the formula alone does not yet answer (e.g., 'What if the piece of watermelon is not a full wedge but has the middle scooped out — how would we find that area?', or 'Can we use this formula if the angle is given in radians instead of degrees?'). Pairs read each other's new question cards and star the one they find most interesting. The teacher collects all formula cards for the DQB's 'What We Now Know' column and posts a selection of starred new question cards in the 'What We Still Wonder' column, reading two aloud to the class.</p> | <p>thinking only. First sticky note: write the sector area formula and explain in your own words what <math>\theta</math> and <math>r</math> mean for our watermelon problem. Second sticky note: write one new question that today's formula has NOT yet answered for you.' WAIT TIME: 4 minutes silent individual work. Then say: 'Read your partner's new question card. Put a star on the one that makes YOU most curious — it can be yours or theirs.' Allow 1 minute. Collect formula cards and post to DQB 'What We Now Know' column. Scan new question cards rapidly and select 2–3 that point toward Lesson 7 content (applying sector and segment area together to a larger design problem). Read these aloud and say: 'These questions are exactly where we are going in Lesson 7, where we deepen our work on segments from Lesson 5. Keep them — they are your intellectual compass.' Post selected new question cards to DQB 'What We Still Wonder' column.</p> | <p>of the class's growing understanding of the driving question. By requiring learners to translate the formula into their own language connected to the phenomenon (Formula Card), the teacher checks for conceptual ownership rather than rote memorisation. The New Question Card makes visible the productive confusion that genuine learning generates, and teacher selection of cards pointing toward Lesson 7 content models scientific inquiry — new knowledge raises new questions — while maintaining narrative continuity in the storyline.</p>  | <p>just abstract symbols) indicate conceptual connection to the phenomenon — teacher counts these as a ratio. Target: <math>\geq 70\%</math> of cards include a phenomenon-linked explanation. (2) New question cards that reference 'arc length' or 'radian' indicate learners whose thinking is running ahead of the sequence — teacher notes these learners for differentiation in Lesson 7. (3) New question cards that re-ask a question already answered (e.g., 'Why does angle affect area?') indicate learners who have not yet consolidated today's key idea — teacher notes these learners for a brief one-to-one check-in at the start of Lesson 7.</p>          |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Radians as ratio of arc length to radius</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Radius or</a></p>                                                                                                                                                                                                                                 | <p>Learners return to the cumulative Sector-Segment Model diagram they have been building since Lesson 1 (a circle with labelled parts, annotated with their growing understanding). Today's task: each learner adds three things to their model — (1) the formalised sector area formula in a labelled box connected by an arrow to the shaded sector region of their diagram; (2) a worked mini-example directly on the diagram using the watermelon context (<math>\theta = 120^\circ</math>, <math>r = 15\text{ cm} \rightarrow A = 75\pi \approx 235.65\text{ cm}^2</math>) annotated with each substitution step; (3) a 'Scaling</p>                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Say: 'Open your cumulative model diagram. Today you are adding the formula, a worked example, and a scaling note — these three additions together answer the core question: WHY does changing only the angle change the area so dramatically? Add them now. You have 12 minutes.' Circulate and prompt: (a) To learners who have written the formula without connecting it to the diagram: 'Draw an arrow from your formula box to the shaded sector. Label the arrow with what the formula calculates.' (b) To learners who have the worked example but no scaling</p>                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Strategy: Cumulative Model Revision with Explicit Scaling Analysis. The cumulative model functions as a personal knowledge-building artefact across all 8 lessons — each addition is not a new diagram but an annotation layer on the same evolving representation. The 'Scaling Note' is the key sensemaking move of this lesson: it asks learners to articulate the <math>r^2</math> relationship in plain language, which directly explains the phenomenon (why a small change in angle or radius makes wedges look dramatically different). This</p> | <p>Observable indicators (Exit Card — 3 minutes at end of Model Building): Teacher writes two questions on the board. Learners answer on a half-sheet of paper to hand in. Q1 (procedural): 'A sector has radius 6 cm and angle <math>150^\circ</math>. Calculate its area correct to 1 decimal place. (Use <math>\pi = 3.142</math>.)' [Expected: <math>A = (150/360) \times 3.142 \times 36 = 47.1\text{ cm}^2</math>] Q2 (conceptual — connects to phenomenon): 'A vendor at Wakulima Market cuts two slices from the same watermelon. Slice A has angle <math>60^\circ</math>, Slice B has angle <math>120^\circ</math>. Without calculating, how many times larger</p> |

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| <p><a href="#">Diameter of a Circle Given Area (Flexbook)</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: CK-12</p> <p><a href="#">Search ARES for similar readings</a></p> | <p>Note' — they shade three different-sized sectors on a small circle sketch and write: 'Doubling the angle doubles the area. Doubling the radius quadruples the area.' This note captures the proportionality insight that explains why the vendor's customers see such dramatically different wedge sizes. Learners who complete all three additions early begin sketching what a segment (sector minus triangle) might look like on their model, in preparation for Lesson 7.</p> | <p>note: 'Now I want you to think about the <math>r^2</math> part. If the vendor doubled the radius of her watermelon from 15 cm to 30 cm, keeping the angle at <math>120^\circ</math>, would the area double? Triple? Quadruple? Write the scaling note.' WAIT TIME: 10 seconds for each prompted learner to think before responding. After 12 minutes, cold-call 2 learners to share their scaling note. Ask the class: 'Does everyone agree that doubling <math>r</math> quadruples the area? Convince me with numbers — someone use the formula for <math>r=15</math> and <math>r=30</math> right now.' WAIT TIME: 12 seconds. Cold-call 1 learner to verify. Spend final 2 minutes on exit card (see Formative Assessment).</p> | <p>connects back to the driving question — 'Why does changing only the angle change the area so dramatically?' — and ensures the formula is understood as a model of a real relationship, not a calculation recipe.</p> | <p>is Slice B than Slice A? Explain your reasoning in one sentence.' [Expected: Slice B is exactly 2 times larger because area is proportional to angle — doubling the angle doubles the area.] Teacher sorts exit cards into three piles after class: (Pile 1) Both correct — ready for Lesson 7 segment work; (Pile 2) Q1 correct, Q2 incorrect — procedural fluency but proportionality concept not yet secure; (Pile 3) Q1 incorrect — formula application errors to address at start of Lesson 7 with a 3-minute re-teach.</p> |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your lesson journal:

1. PHENOMENON CONNECTION: In how many of the five phases did you explicitly name the Wakulima Market watermelon vendor? If fewer than four, identify which phase lost the phenomenological thread and plan how to restore it in Lesson 7.
2. FORMULA OWNERSHIP: When you cold-called learners to explain the formula after the derivation, what language did they use? Did learners say ' $\theta$  over 360 is the fraction of the circle' (conceptual) or simply 'you plug in the numbers' (procedural)? How will you deepen conceptual language in Lesson 7's segment derivation?
3. GRADUATED EXAMPLES: Which of the three examples (watermelon, gate, irrigation pivot) caused the most difficulty? If the unit conversion in Example 3 ( $m^2$  to hectares) caused widespread errors, consider whether Lesson 7 should open with a brief unit-conversion warm-up before introducing the segment formula.
4. DQB QUALITY: Count the proportion of New Question Cards that referenced 'segment', 'chord', or 'arc length' — this is your leading indicator of readiness for Lesson 7. If fewer than 30% of learners generated such questions, plan a brief 'Look at this shape' hook at the start of Lesson 7 to explicitly create the need for segment area.
5. EXIT CARD DATA: Use the three exit-card piles to inform Lesson 7 groupings. Consider pairing a Pile 1 learner with a Pile 2 learner during Lesson 7's segment derivation activity, and plan a 3-minute whole-class re-entry activity for Pile 3 learners before the main lesson begins.
6. PACING: This lesson targets 80 minutes. If the Explain phase (Examples 1–3) ran beyond 35 minutes, identify which example consumed excess time and whether learners needed more scaffolding on the  $270^\circ$  reflex angle or the  $r^2$  scaling idea — both are critical for Lesson 7's segment work.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| <p><b>What did I observe?</b></p> | <p>Learners observed a teacher-led step-by-step recap of the derivation of the sector area formula <math>A = (\theta/360) \times \pi r^2</math> from first principles, built on the Wakulima Market watermelon phenomenon. They also observed three real-life Kenyan sector contexts: a <math>120^\circ</math> watermelon slice (<math>r = 15</math> cm), a school gate sweep (<math>r = 1.2</math> m, <math>\theta = 75^\circ</math>), and a Naivasha centre-pivot irrigation arc (<math>r = 80</math> m, <math>\theta</math></p> |
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|                                              | = 270°). Learners observed peer pair presentations on mini-whiteboards and whole-class evaluation of solution accuracy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>What did I learn?</b>                     | Learners learned that the area of any sector is a proportional fraction of the full circle's area, expressed as $A = (\theta/360) \times \pi r^2$ . They learned that sector area scales linearly with angle $\theta$ (doubling the angle doubles the area) and quadratically with radius $r$ (doubling the radius quadruples the area). They applied this formula to calculate areas in cm <sup>2</sup> , m <sup>2</sup> , and hectares across three graduated Kenyan contexts, and updated the class Driving Question Board with the formalised formula and new arising questions pointing toward applying sector and segment area together in Lesson 7.                           |
| <b>How does this explain the phenomenon?</b> | Learners explained in their cumulative model diagrams and DQB Formula Cards why changing only the angle of a cut at Wakulima Market changes the area of the watermelon wedge so dramatically: because sector area is directly proportional to the central angle ( $A \propto \theta$ ), a vendor who cuts at 120° instead of 60° produces a slice exactly twice as large — resolving the customers' complaint with mathematical precision. Learners also explained the $r^2$ relationship in a written Scaling Note, and exit card responses documented both procedural fluency (correct substitution) and conceptual understanding (proportionality reasoning without calculation). |

## LESSON 7 (80 minutes): Investigating Segment Area: From Sector to Slice — Unlocking the Curved Piece

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <b>Purpose</b>              | Learners will determine the area of a segment of a circle by subtracting the area of a triangle from the area of a sector, and apply this to solve fair-sharing problems rooted in the Wakulima Market watermelon phenomenon, where the segment is the piece bounded by a chord and the curved rind arc.                                                                                                                                                                                                                                                                            |
| <b>Knowledge</b>            | Learners will know that: (a) a segment is the region between a chord and the arc it subtends; (b) the area of a segment = area of sector – area of triangle (Area = $\frac{1}{2}r^2\theta - \frac{1}{2}r^2\sin\theta$ , where $\theta$ is in radians, OR Area = $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ , where $\theta$ is in degrees); (c) the triangle formed by two radii and a chord has area $\frac{1}{2}r^2\sin\theta$ .                                                                                                                                            |
| <b>Skills</b>               | Learners will be able to: (a) identify and distinguish a segment from a sector in a diagram; (b) calculate the area of the triangle formed by two radii and a chord using the formula $\frac{1}{2}r^2\sin\theta$ ; (c) compute the area of a segment by subtracting the triangle area from the sector area; (d) apply segment area to real-life contexts such as watermelon slices and design problems.                                                                                                                                                                             |
| <b>Attitudes</b>            | Learners appreciate how precise mathematical reasoning — rather than visual estimation — is the only reliable tool for ensuring fairness in real-life sharing situations. Learners display Unity by working collaboratively on cut-out investigations, and Peace by tolerating different approaches to solving segment problems.                                                                                                                                                                                                                                                    |
| <b>Key Inquiry Question</b> | Where do we use area of part of a circle in real life?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Purpose in Storyline</b> | In Lessons 2–6, learners derived and practised sector area and explored how changing the angle of a cut changes the wedge size. But the mama mboga at Wakulima Market sometimes cuts straight across a watermelon wedge along a chord, leaving a piece bounded by a flat chord and the curved rind arc: a segment. This lesson investigates: how much watermelon (segment area) does a customer actually receive in that rind-bounded piece? This is the key missing piece in the evidence sequence — learners now have all the tools needed to solve the full fair-sharing puzzle. |
| <b>Safety Notes</b>         | If physical cutting tools (scissors, rulers) are used for paper cut-outs, learners should handle them carefully and place them flat on the desk when not in use. No sharp knives or actual watermelons are needed; paper or card cut-outs are sufficient.                                                                                                                                                                                                                                                                                                                           |

### B. LESSON OVERVIEW

Lesson 7 is the penultimate lesson in the 8-lesson sequence on Area of Part of a Circle (Sub-Strand 2.5). Learners derived the sector area formula in Lesson 2 and met segments and the  $\frac{1}{2}r^2\sin\theta$  triangle formula in Lesson 5. In this lesson, they investigate the area of a circular segment — the region between a chord and its arc — using a cut-out decomposition activity: separating a sector into a triangle and a segment, measuring, calculating, and reconciling results. The anchoring phenomenon (the mama mboga at Wakulima Market cutting watermelon) is deepened: when a wedge is cut straight across a chord, the rind-bounded piece is a segment, and fair pricing demands its exact area. By the end of the lesson, learners can compute segment area using  $\text{Area}(\text{segment}) = \text{Area}(\text{sector}) - \text{Area}(\text{triangle}) = (\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ , and they update both their Driving Question Board and their cumulative circle-area model.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of Sectors and Segments Principles - Basic</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Area of Sectors and Segments (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p>           | <p>Learners are shown a large diagram on the board: a circle of radius 14 cm with a chord drawn across it, shading only the smaller region between the chord and the arc (a minor segment). The teacher reminds learners of the mama mboga scenario: 'She cuts a watermelon wedge straight across a chord. The customer receives the rind-bounded piece — this shaded region.' Learners individually write a prediction in their exercise books: (a) Is the shaded segment larger or smaller than the sector with the same angle? (b) Estimate the area of the segment if the sector angle is <math>60^\circ</math> and <math>r = 14</math> cm. They share predictions with a shoulder partner (Think-Pair-Share, 2 minutes). Three pairs are cold-called to share their estimates.</p> | <p>Display the diagram using a poster or drawn clearly on the board. Say: 'Look carefully at the shaded region. This is the rind-bounded piece of watermelon the customer receives after the wedge is cut straight across a chord. In your books, write: is this region bigger or smaller than the full sector wedge? Then estimate its area — your best mathematical guess — if <math>r = 14</math> cm and the sector angle is <math>60^\circ</math>. You have 3 minutes working alone.' [WAIT 3 minutes.] 'Now share with your shoulder partner. Do you agree?' [WAIT 1 minute.] Cold-call exactly 3 pairs using the register: 'Pair 4 — Akinyi and Otieno — what did you predict and why?' After each response, probe: 'What formula or reasoning did you use to arrive at that estimate?' [WAIT 10 seconds after each probe before accepting responses.] Record the range of estimates on the board without confirming or denying correctness: 'We have predictions from ____ <math>\text{cm}^2</math> to ____ <math>\text{cm}^2</math>. Let us find out who is closest and why.'</p> | <p>Think-Pair-Share with anchored estimation. This activates learners' prior knowledge of sector area (Lessons 4–6) and surfaces the key misconception that a segment is simply a smaller sector. By estimating before calculating, learners create cognitive dissonance that motivates the observation phase.</p>                                                                                                                                                                       | <p>Observable indicator: learners write a numerical estimate (not just 'smaller'). Listen for whether learners attempt to use the sector formula — this reveals whether sector area from Lessons 2–6 has been consolidated. Flag any learner who writes 'area = arc length <math>\times r</math>' — this confuses arc and area and needs targeted correction in Phase 3.</p>                                                                                                                                                                                                        |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of Sectors and Segments Principles - Basic</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Area of Sectors and Segments Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Activity 1 — Cut-Out Decomposition (30 min, groups of 4): Each group receives a pre-drawn circle of radius 14 cm on card, with a sector of <math>60^\circ</math> clearly marked and a chord connecting the two radii endpoints. Learners label the parts: sector OAB, triangle OAB, and segment AB (the region between chord AB and arc AB). Step 1: Learners cut out the full sector. Step 2: They fold/cut along the chord to separate the triangle from the segment. Step 3: They calculate the sector area using <math>(\theta/360)\pi r^2 = (60/360) \times \pi \times 14^2 \approx 102.6 \text{ cm}^2</math>.</p>                                                                                                                                                              | <p>Distribute materials (pre-drawn card circles, scissors, rulers, calculators) and say: 'Today you will physically decompose a sector into two parts: a triangle and a segment. Your job is to find the area of just the curved piece — the rind-bounded segment of the watermelon, left after the triangle is removed.' Walk group to group every 4 minutes. At each group, ask: 'Point to the segment. Now point to the triangle. Which is bigger?' [WAIT 10 seconds.] If a group is stuck on the triangle area, prompt: 'We introduced the area of a triangle from two radii and an</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Concrete decomposition (cut-and-separate). Making the abstract formula tactile — physically separating the triangle from the segment — gives learners a concrete referent for the operation 'sector minus triangle.' The shared class data table introduces co-construction of evidence, a key NGSS science and engineering practice (Analysing and Interpreting Data). Learners see that the segment grows nonlinearly with angle because <math>\sin\theta</math> is not linear.</p> | <p>Observable indicators: (a) Every group can correctly label the segment vs. the sector vs. the triangle on their cut-out (check by circulating). (b) Every learner records a numeric segment area for at least one angle in the data table. (c) Listen for the phrase 'sector minus triangle' — if groups say 'sector minus arc,' intervene immediately: 'An arc is a length, not an area. What shape are you subtracting?' Red flag: groups who subtract arc length from sector area. These learners need the Phase 3 explicit instruction most urgently — note their names.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Step 4: They calculate the area of triangle OAB using <math>\frac{1}{2}r^2\sin\theta = \frac{1}{2} \times 14^2 \times \sin 60^\circ \approx 84.9 \text{ cm}^2</math>. Step 5: They find segment area = <math>102.6 - 84.9 = 17.7 \text{ cm}^2</math>. Step 6: They physically place the triangle cut-out on the segment to verify that the triangle is larger. Activity 2 — Data Table (10 min): Each group completes a table for three different angles (<math>60^\circ</math>, <math>90^\circ</math>, <math>120^\circ</math>) at <math>r = 14 \text{ cm}</math>, recording: Sector Area, Triangle Area, Segment Area. Groups record results on a shared class table drawn on the board.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>included angle in Lesson 5 (from Sub-Strand 2.4). What was that formula?' [WAIT 15 seconds — do NOT supply the answer immediately.] When groups finish Step 5, ask: 'Does your segment area match your Phase 1 prediction? Why or why not?' [WAIT 10 seconds.] For the class data table, cold-call one representative from 4 different groups to fill in one row each. Say: 'Amina from Group 3, please write your <math>90^\circ</math> results on the board.' After all rows are filled, ask the class: 'What pattern do you notice as the angle increases? Does the segment always grow at the same rate as the sector?' [WAIT 15 seconds before cold-calling 2 learners.]</p>                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of a circle intuition</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Area of Sectors and Segments: Ms. Arcade-Man (PLIX)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to whole class. The teacher derives the segment area formula explicitly on the board, connecting each step to the cut-out. Learners copy into their books and then solve two worked examples independently before a peer-check. Worked Example 1 (Wakulima context): A watermelon of radius 21 cm is cut with a chord so that the sector angle is <math>90^\circ</math>. Calculate the area of the segment (the red flesh). Answer: Sector = <math>(90/360) \times \pi \times 21^2 = 346.4 \text{ cm}^2</math>; Triangle = <math>\frac{1}{2} \times 21^2 \times \sin 90^\circ = \frac{1}{2} \times 441 \times 1 = 220.5 \text{ cm}^2</math>; Segment = <math>346.4 - 220.5 = 125.9 \text{ cm}^2</math>. Worked Example 2 (Design context): A circular window of radius 10 cm has a decorative segment with a <math>120^\circ</math> arc. Find the segment area. Answer: Sector = <math>(120/360) \times \pi \times 100 = 104.7 \text{ cm}^2</math>; Triangle = <math>\frac{1}{2} \times 100 \times \sin 120^\circ = 50 \times 0.866 = 43.3 \text{ cm}^2</math>; Segment = <math>104.7 - 43.3 = 61.4 \text{ cm}^2</math>. Learners</p> | <p>Begin by connecting to the cut-out: 'We physically separated the sector into a triangle and a segment. The formula simply does that subtraction mathematically.' Write on the board step by step: 'Area of segment = Area of sector – Area of triangle.' Then: 'Area of sector = <math>(\theta/360) \times \pi \times r^2</math>.' Then, recalling Lesson 5: 'Area of triangle OAB = <math>\frac{1}{2} \times r \times r \times \sin\theta = \frac{1}{2}r^2\sin\theta</math>.' Then: 'Therefore: Area of segment = <math>(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta</math>.' Box this formula. Say: 'This is the core formula for today. Write it, box it, and label every variable.' [WAIT 1 minute for all learners to write.] Work through Example 1 aloud, pausing at each substitution: 'What is <math>\theta</math> here? What is <math>r</math>? What do I substitute into the sector formula first?' [WAIT 10 seconds after each question before cold-calling — cold-call 3 different learners across Example 1.] For Example 2, do the sector step only, then say: 'You</p> | <p>Explicit formula derivation anchored to the physical model (NGSS: Developing and Using Models). By connecting the algebraic formula to the physical cut-out, the teacher bridges concrete and abstract representations. Peer-marking builds metacognitive awareness and communication skills (KICD Core Competency: Communication and Collaboration). The Kenyan food contexts (watermelon, chapati) maintain the phenomenon thread throughout.</p> | <p>Observable indicators: (a) Every learner's book contains the boxed formula with all variables labelled. (b) During the independent chapati problem, circulate and check that learners are using <math>\frac{1}{2}r^2\sin\theta</math> for the triangle (not <math>\frac{1}{2} \times \text{base} \times \text{height}</math>, which requires additional steps). (c) During peer-marking, listen for disagreements — these reveal common errors (e.g., forgetting to convert degrees to the correct trig input, or subtracting sector from triangle instead of the reverse). Cold-call one pair where the marker identified an error and ask: 'What mistake did you spot and how should it be corrected?'</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | then solve a third problem independently: a chapati of radius 12 cm is cut with a chord at $80^\circ$ . Find the segment area. After 5 minutes, they swap books with a partner for peer-marking using the board solution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | complete the triangle step and the subtraction — 2 minutes, alone.' [WAIT 2 minutes.] Cold-call 2 learners to give the triangle area and the final answer. For the independent chapati problem, say: 'Work entirely alone. Show every step. You have 5 minutes.' [WAIT 5 minutes strictly.] Then: 'Swap books. Mark your partner's work as I reveal the solution. Write one thing they did well and one thing to improve.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">Area of a sector</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Derivation of the Triangle Area Formula Practice</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | Learners return to the class Driving Question Board (DQB) displayed at the front. The DQB has been building since Lesson 1, tracking what learners know, what they wonder, and what evidence they have gathered. In pairs, learners write on sticky notes or designated DQB cards: (a) ONE new thing they can now explain about the watermelon phenomenon that they could not explain before this lesson; (b) ONE remaining question they still have about the fair-sharing problem. Pairs post their cards under the appropriate DQB columns ('Now I Can Explain...' and 'I Still Wonder...'). The teacher reads 3–4 cards aloud and the class votes on which remaining questions are most important to resolve in Lesson 8 (the final lesson). | Point to the DQB and say: 'Look at our board. At the start of this unit, we asked: how can we calculate the exact area of any wedge or segment of a circle, like the mama mboga's watermelon slices, so every customer gets a fair share? Today we unlocked the segment. Write on your card: one thing you can now explain about the phenomenon that you could not explain before.' [WAIT 2 minutes.] 'Now write your remaining question — what do you still need to know to fully solve the fair-sharing problem?' [WAIT 1 minute.] Ask pairs to post cards. Read 3–4 aloud, then say: 'Raise your hand if your remaining question is about applying both sector AND segment area together to solve a real problem.' Acknowledge hands: 'That is exactly what Lesson 8 will address. We are almost at the full answer to our driving question.' Highlight any DQB card that mentions comparing a segment and a sector from the same circle — this is the conceptual bridge to Lesson 8. | DQB Update (metacognitive reflection and collective knowledge tracking). This strategy, used consistently across the 8-lesson sequence, makes learning visible and helps learners see their own progress along the storyline. Voting on remaining questions gives learners agency and primes them for Lesson 8. It also surfaces any persistent misconceptions (e.g., confusing segment and sector) that the teacher must address before the final lesson. | Observable indicators: (a) Every pair posts at least one 'Now I Can Explain' card that references segment area specifically (not just 'area of a circle'). (b) Teacher reads 'I Still Wonder' cards to check for productive questions (e.g., 'What if the angle is more than $180^\circ$ ?' or 'How do I find segment area without a calculator for $\sin\theta$ ?') vs. unproductive confusion (e.g., 'I don't understand anything about segments'). Learners with unproductive confusion cards are identified for targeted support before Lesson 8. |
| <b>Model Building Phase</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Learners open their cumulative circle-area model booklet (a folded A3 sheet that has been updated in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Say: 'Open your model booklets to the next blank page. We are going to add the segment to our growing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Cumulative model revision (NGSS: Developing and Using Models). The model booklet is the physical                                                                                                                                                                                                                                                                                                                                                           | Observable indicators: (a) Every learner's model diagram correctly distinguishes the segment                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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| <p> <b>VIDEO:</b><br/> <a href="#">Area of a circle</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Area of Sectors and Segments (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>each lesson). In Lessons 1–6, they drew and annotated: a full circle (area = <math>\pi r^2</math>), a sector (area = <math>(\theta/360)\pi r^2</math>), and the relationship between angle and sector area. Today they add: a clearly labelled diagram of a circle showing a chord, a segment (shaded), the triangle OAB (unshaded), and the sector OAB (the sum). They write the segment area formula beside the diagram and annotate each component: 'This part is the sector,' 'This part is the triangle I subtract,' 'This shaded part is the segment — the rind-bounded piece.' They also add a second diagram: the Wakulima Market watermelon slice, labelled with <math>r = 21</math> cm and <math>\theta = 90^\circ</math>, with the calculated segment area (<math>125.9</math> cm<sup>2</sup>) written on the shaded region. Finally, learners write one sentence connecting today's model to the driving question: 'Now I can calculate the exact area of the rind-bounded piece in any watermelon segment, so I can check whether two pieces are truly equal even if one looks bigger.'</p> | <p>model of circle areas. First, draw a circle, mark the centre O, draw two radii OA and OB at <math>90^\circ</math>, then draw chord AB. Shade only the segment — the region between AB and the arc. Label: sector OAB, triangle OAB, segment AB.' [WAIT 3 minutes for drawing.] Circulate and check: 'Show me which part you are shading. Is that the segment or the sector?' [WAIT 10 seconds per group.] 'Now write the formula beside your diagram — box it.' [WAIT 1 minute.] 'Add the Wakulima watermelon diagram with <math>r = 21</math> cm, <math>\theta = 90^\circ</math>, and the segment area we calculated: <math>125.9</math> cm<sup>2</sup>.' [WAIT 2 minutes.] 'Finally, write one sentence: how does today's model help you answer the driving question?' Cold-call 2 learners to read their sentence aloud. Affirm and connect: 'Excellent. In Lesson 8 — our final evidence lesson — you will use BOTH the sector formula and the segment formula together to solve the complete fair-sharing problem for the mama mboga. Your model is almost complete.'</p> | <p>artefact of the entire 8-lesson storyline. Each revision makes the growing understanding visible and honours prior work. Writing a connecting sentence to the driving question is a disciplinary literacy move that consolidates conceptual understanding and prepares learners for the summative application in Lesson 8.</p> | <p>(shaded) from the triangle (unshaded) and from the full sector. (b) The formula is written correctly and boxed. (c) The connecting sentence explicitly references the driving question ('fair share,' 'exact area,' 'watermelon,' or equivalent). Collect 5 model booklets (selected randomly) to review overnight and provide written feedback before Lesson 8.</p> |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Could all learners correctly identify the segment (not the sector) in the cut-out activity? If not, which groups confused the two, and how will you address this at the start of Lesson 8? (2) Did learners fluently apply the triangle area formula  $\frac{1}{2}r^2\sin\theta$  from Lesson 5 (Sub-Strand 2.4), or did the prerequisite knowledge create a bottleneck? If many learners struggled here, plan a 5-minute review of  $\frac{1}{2}r^2\sin\theta$  at the start of Lesson 8 before introducing combined applications. (3) Were the DQB 'I Still Wonder' cards mathematically productive? Do learners' remaining questions point toward the combined sector-and-segment application that Lesson 8 requires, or are there fundamental gaps that need to be closed first? (4) Review the 5 model booklets collected: are the diagrams correctly labelled and the formulas correctly written? Use findings to adjust the first 10 minutes of Lesson 8. (5) Did the Kenyan food contexts (watermelon at Wakulima Market, chapati) keep learners engaged and grounded in the phenomenon, or did the mathematical complexity cause some learners to lose the contextual thread? Plan to re-anchor the phenomenon explicitly at the opening of Lesson 8.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>What did I observe?</b>                   | Learners cut out a sector of radius 14 cm and 60°, separated it into a triangle (OAB) and a segment, and calculated each area. They also completed a class data table for angles 60°, 90°, and 120°, recording sector area, triangle area, and segment area for each.                                                                                                                               |
| <b>What did I learn?</b>                     | The area of a segment = area of sector – area of triangle = $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ . The triangle formed by two radii and a chord has area $\frac{1}{2}r^2\sin\theta$ . A segment is always smaller than its corresponding sector because the triangle area is always positive.                                                                                           |
| <b>How does this explain the phenomenon?</b> | For the mama mboga's watermelon at Wakulima Market: when a wedge is cut straight across a chord, the customer receives a segment — the rind-bounded piece between the chord (cut face) and the curved rind arc. Using the segment formula, the exact area of this piece in a 90° slice of radius 21 cm is 125.9 cm². This means we can now price each piece fairly by area, not by visual estimate. |

## LESSON 8 (80 minutes): Segment Area, Final Explanation & Fair-Share Verdict — Completing the Watermelon Puzzle

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>Purpose</b>              | To determine the area of a segment of a circle and apply the area of part of a circle to real-life fair-sharing and design problems, culminating in a full explanation of the anchoring phenomenon.                                                                                                                                                                                                                                                                                                                                                                |
| <b>Knowledge</b>            | Learners will know that the area of a circular segment equals the area of the corresponding sector minus the area of the triangle formed by the two radii and the chord; expressed as: Area of segment = $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ .                                                                                                                                                                                                                                                                                                        |
| <b>Skills</b>               | Learners will be able to: (a) identify and isolate the triangle inside a sector using a chord; (b) calculate the area of a triangle using $\frac{1}{2}r^2\sin\theta$ ; (c) subtract triangle area from sector area to find segment area; (d) apply both sector and segment formulae to solve fair-sharing and design problems; (e) construct and compare cumulative circle-area models from Lesson 1 through Lesson 8.                                                                                                                                             |
| <b>Attitudes</b>            | Learners will appreciate how precise mathematical calculation — rather than visual guessing — guarantees fairness in real-life sharing situations, and will demonstrate confidence in presenting a complete, evidence-based explanation to peers.                                                                                                                                                                                                                                                                                                                  |
| <b>Key Inquiry Question</b> | Where do we use area of part of a circle in real life?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Purpose in Storyline</b> | In this final lesson learners work with new sector cut-outs ( $r = 18$ cm). By cutting out the triangle (formed by the two radii and the chord) from each sector, they isolate the segment and consolidate its area formula from Lessons 5 and 7. They then use both sector and segment formulae to adjudicate the Wakulima Market watermelon dispute — calculating exactly how much watermelon (in $\text{cm}^2$ ) each customer received — and revise their cumulative model one last time to produce a complete, evidence-based answer to the driving question. |
| <b>Safety Notes</b>         | Scissors and cutting tools are to be handled with care; learners must cut away from their bodies and keep blades flat on the desk when not in use. Scrap paper pieces must be collected and disposed of responsibly to keep the classroom environment clean.                                                                                                                                                                                                                                                                                                       |

### B. LESSON OVERVIEW

Lesson 8 is the capstone lesson for Sub-Strand 2.5. Learners begin with a Predict phase in which they estimate how much area is 'lost' when the triangular portion is removed from a sector to leave only the curved segment. In the Observe phase they physically cut the triangle from their sector cut-outs, measure dimensions, and record data. In the Explain phase they derive Area of segment =  $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ , apply it to the Wakulima Market watermelon scenario (given realistic values:  $r = 18$  cm,  $\theta$  values of  $40^\circ$ ,  $65^\circ$ ,  $90^\circ$ , and  $120^\circ$ ), and produce a fair-share verdict supported by calculations. The DQB phase marks all remaining questions as answered and surfaces the complete causal story. The Model Building phase invites learners to compare their Lesson 1 naïve drawing to their final annotated model, articulating every conceptual addition made across the unit. The lesson closes with a whole-class Final Explanation Gallery Walk and structured teacher-led consolidation.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                        | Learner Experience                                                                                                                   | Teacher Moves                                                                                     | Sensemaking Strategy                                                                           | Formative Assessment Strategy                                                                 |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Area of Sectors</a> | Each group receives sector cut-outs of radius 18 cm with $\theta = 40^\circ$ , $65^\circ$ , $90^\circ$ and $120^\circ$ . The teacher | Teacher opens by pointing to the projected watermelon image and saying: 'Last lesson we found the | Prediction Journaling with Public Commitment: by writing and then publicly sharing a numerical | Observable indicator: every learner has a written numerical estimate (fraction or percentage) |

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| <p><a href="#">and Segments Principles - Basic</a><br/>Source: CK-12<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/><a href="#">Area of Sectors and Segments (Flexbook)</a><br/>Source: CK-12<br/><a href="#">Search ARES for similar readings</a></p>                                                                                                                                                  | <p>displays a large projected image of a watermelon sector with a straight chord drawn across it — separating the triangular piece (two radii + chord) from the curved segment (the rind-bounded sliver). Learners record individual written predictions in their science notebooks responding to two prompts: (1) 'Estimate: what fraction of the sector's area do you think the segment (the curved sliver beyond the chord) represents?' and (2) 'Will a wide-angle segment (<math>\theta = 120^\circ</math>) always have a larger area than a narrow-angle segment (<math>\theta = 40^\circ</math>) of the same radius? Explain your reasoning.' Groups share predictions via a quick show-of-hands poll on the fraction estimate, and the teacher records the spread of estimates on the board without confirming or correcting.</p>                                                                                                                                                                 | <p>area of the full wedge — the sector. But look at this piece [points to the curved sliver beyond the chord]. A customer at Wakulima Market was sold ONLY this curved piece — no triangle. She says she was cheated. Was she? We need exact numbers.' Pause 10 seconds. Teacher writes both prediction prompts on the board and gives learners 3 minutes of silent individual thinking time. Teacher circulates, reading notebooks — do NOT give hints. After 3 minutes, cold-call 4 learners (one per group) for their fraction estimate. Record responses verbatim (e.g., '<math>\frac{1}{4}</math> of the sector', '<math>\frac{1}{2}</math>', 'about 30%') in a T-chart on the board labelled 'Our Predictions'. Teacher then says: 'Keep these predictions visible — we will return to them at the end of Phase 2 to see who was closest.'</p>                                | <p>estimate before any instruction, learners activate prior knowledge of sector area and create cognitive dissonance when they realise they lack a method to handle the curved boundary — directly mirroring the puzzle the Wakulima Market customer faces. This primes them to attend purposefully to the subtraction strategy in Phase 2.</p>                                                                                                                                                                                                                                               | <p>and a written sentence of reasoning in their notebook before Phase 2 begins. Teacher scans for misconceptions — particularly learners who write 'segment area = <math>\frac{1}{2} \times</math> sector area regardless of <math>\theta</math>', flagging those groups for targeted questioning during Phase 2.</p>                                                                                                                                                                                                                            |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/><a href="#">Area of Sectors and Segments Principles - Basic</a><br/>Source: CK-12<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/><a href="#">Area of Sectors and Segments Practice</a><br/>Source: CK-12<br/><a href="#">Search ARES for similar readings</a></p> | <p>Each group receives: (a) their labelled sector cut-outs (<math>r = 18</math> cm); (b) a ruler and a pencil; (c) a printed Group Data Table (columns: <math>\theta^\circ</math>, <math>r</math> cm, chord length <math>c</math> cm, sector area <math>\text{cm}^2</math>, triangle area <math>\frac{1}{2}r^2\sin\theta</math> <math>\text{cm}^2</math>, segment area <math>\text{cm}^2</math>). Step 1 (5 min): Using a ruler, learners draw the chord connecting the two ends of the arc on their sector cut-out, then cut along the chord to physically separate the triangular piece from the curved segment. Both pieces are labelled and kept. Step 2 (5 min): Learners measure the chord length <math>c</math>, verify the radius <math>r = 18</math> cm, and record. Step 3 (5 min): Groups calculate the area of the triangle using the formula <math>\frac{1}{2}r^2\sin\theta</math> (introduced in Sub-Strand 2.4 Area of Polygons — teacher reminds learners this is the 'two-sides-and-</p> | <p>Before learners begin cutting, teacher models the procedure ONCE using a large demonstration cut-out on the board: 'I draw my chord here — joining the two points where the radii meet the arc. I cut along the chord. Now I have two pieces: the triangle [holds up triangle] and the segment [holds up curved sliver]. I want the area of THIS piece [holds up segment].' Pause 10 seconds. 'How can I get it if I already know the sector area?' Wait 12 seconds for think time, then cold-call 2 learners. Guide toward the subtraction idea without stating it directly. Circulate during group work. At the 8-minute mark, stop all groups briefly: 'Check: has everyone used <math>\frac{1}{2}r^2\sin\theta</math> for the triangle — not <math>\frac{1}{2} \times</math> base <math>\times</math> height? Why do we use <math>\frac{1}{2}r^2\sin\theta</math> here?'</p> | <p>Concrete Manipulation with Data Recording: physically cutting the triangle from the sector makes the subtraction relationship tangible and irreversible in learners' minds. Recording both pieces' areas in the same table column-by-column scaffolds the algebraic relationship (Segment = Sector – Triangle) as a pattern across four different angle values, not merely a one-off formula. This connects directly to the phenomenon: each watermelon slice the vendor cuts has both a sector area and a triangular-central portion, and the curved rind piece alone is the segment.</p> | <p>Observable indicator: each group's Data Table has all six columns populated with correct units (<math>\text{cm}^2</math>) for all four <math>\theta</math> values. Teacher checks specifically that the triangle area column uses <math>\frac{1}{2}r^2\sin\theta</math> (not <math>\frac{1}{2} \times</math> base <math>\times</math> height from a measurement), and that segment area values are smaller than sector area values — a necessary logical check. Groups whose segment area exceeds sector area are immediately redirected.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | included-angle' formula they already know). They record sector area (calculated with the sector formula from Lesson 2) and compute: Segment area = Sector area – Triangle area. Step 4 (3 min): Groups compare their computed segment area to their original prediction from Phase 1 and note the difference in their notebooks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Cold-call 1 learner. At the 13-minute mark prompt groups: 'Compare your answer to your Phase 1 prediction. How far off were you? Write one sentence about WHY you were off or right.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area of a circle</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Area of a Sector (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Part A — Formula Derivation (8 min): Each group writes the general formula on a large sheet of manila paper: Area of segment = <math>(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta</math>. They factor it as <math>\frac{1}{2}r^2[(\theta\pi/180) - \sin\theta]</math> and annotate each term in a different colour, labelling what it represents physically (sector portion vs. triangle portion). They verify the formula using their own measured values (<math>r = 18</math> cm, <math>\theta = 90^\circ</math>) and confirm it matches their cut-and-subtract result. Part B — Wakulima Market Verdict (10 min): Groups receive the 'Customer Complaint Card': 'Mama Wanjiku at Wakulima Market cut a circular watermelon (<math>r = 18</math> cm) into four unequal wedge-shaped pieces. The central angles were: Piece A = <math>40^\circ</math>, Piece B = <math>65^\circ</math>, Piece C = <math>90^\circ</math>, Piece D = <math>120^\circ</math>. She then cut a chord across each piece and sold ONLY the curved segment to each customer, keeping the triangular centres to make juice. Customer Ochieng (Piece D) complained that Customer Auma (Piece C) received a larger segment even though Ochieng's angle was bigger. Was Ochieng right?' Groups calculate all four segment areas and write a one-paragraph verdict with supporting calculations. Part C —</p> | <p>For Part A, teacher writes on the board: 'Sector area = <math>(\theta/360)\pi r^2</math>. Triangle area = <math>\frac{1}{2}r^2\sin\theta</math>. Therefore, Segment area = ?' and waits 10 seconds before taking responses. Cold-call 2 learners to complete the subtraction. Teacher then says: 'Factor out <math>\frac{1}{2}r^2</math> from both terms. What do you get?' Wait 12 seconds. Cold-call 1 learner. Write the factored form alongside the expanded form — tell learners both are correct and acceptable in assessments. For Part B, circulate and prompt groups that are stuck: 'First calculate sector area for Piece D. Then calculate triangle area for Piece D. Now subtract. Does the result surprise you?' After 10 minutes, cold-call one group to present their verdict. If the verdict is correct (Ochieng is partially right — Piece D segment IS larger than Piece C segment), ask: 'Why did the segment areas NOT increase in the same proportion as the angles? What role does <math>\sin\theta</math> play?' Wait 15 seconds — this is a high-cognitive-demand question. Accept reasoning rather than a precise answer. For Part C, cold-call 2 different groups for their Kenyan contexts. Explicitly connect each back to the driving question: 'So the answer to our driving question</p> | <p>Claim-Evidence-Reasoning (CER) Writing: the Customer Complaint Card requires learners to state a claim (Ochieng is right/wrong), support it with numerical evidence (four segment area calculations), and explain their reasoning (relationship between <math>\theta</math>, <math>\sin\theta</math>, and segment area). This is a full scientific argument structure applied to a mathematics context, directly answering the driving question with real numbers from the phenomenon.</p> | <p>Observable indicator: each group's manila paper shows (1) the formula with colour-coded annotations, (2) a numerical verification using <math>r = 18</math> cm and their own <math>\theta</math>, and (3) a written verdict paragraph with all four segment areas calculated to 1 decimal place. Teacher listens for learners correctly interpreting why <math>\sin\theta</math> grows more slowly than <math>\theta</math> for large angles — this nuanced reasoning distinguishes 'Meets Expectations' from 'Exceeds Expectations' on the KICD rubric.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Connections (4 min): Groups identify two additional real-life contexts in Kenya where segment area matters (suggested: the area of land irrigated by a pivot-irrigation system beyond a fence line in Rift Valley; the area of a decorative tile pattern at a Mombasa hotel).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | now includes segments, not just sectors.'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                            |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">Area of a sector</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Area of a Sector Practice</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | The full Driving Question Board (DQB), built incrementally across all 8 lessons, is displayed at the front of the room. Each group receives three sticky-note colours: GREEN = 'We can now answer this fully', YELLOW = 'We can answer this partially', RED = 'Still puzzling us'. Groups spend 4 minutes silently reviewing every card on the DQB and deciding on a colour for each question their group originally posted. Groups then send one representative to the board to re-label cards. The class reviews any remaining RED cards (if any) and attempts a brief collective answer. The teacher reads the original Driving Question aloud one final time: 'How can we calculate the exact area of any wedge-shaped or segment-shaped piece of a circle — like a watermelon slice at Wakulima Market — so that we can solve fair-sharing and design problems in real life?' The whole class co-constructs a one-sentence Final Answer on the board, and every learner copies it into their notebook. | Teacher points to the DQB and says: 'This board has been our map across 8 lessons. Let's see how much of it we can now colour green.' Give groups 4 minutes of quiet review time — do not answer questions during this time; redirect all queries back to group discussion. After 4 minutes, invite representatives to re-label cards. Stand back and observe — do not direct placement. After all groups have labelled, scan the board and say: 'I see [X] green cards, [Y] yellow cards, [Z] red cards. Let's look at any remaining red cards together.' If red cards remain (expected: at most 1–2 extension-level questions), facilitate a 2-minute class discussion. Then ask: 'In one sentence — using the words RADIUS, ANGLE, SECTOR, SEGMENT, and FORMULA — can someone complete this starter: The exact area of any wedge or segment of a circle can be calculated by...?' Wait 12 seconds. Cold-call 1 learner, then invite additions from 2 more peers. Write the agreed sentence on the board in a box titled 'OUR FINAL ANSWER'. Learners copy it. | DQB Audit with Colour Coding: revisiting every question posted throughout the unit with a traffic-light system makes the progression of understanding visible and metacognitive. Learners experience closure — a rare and motivating feeling in mathematics — while the constraint of using five specific vocabulary words in the Final Answer forces precise scientific language rather than vague summary. | Observable indicator: at least 80% of DQB cards are marked GREEN by the end of Phase 4. Teacher records the specific cards marked YELLOW or RED as evidence of residual gaps to address in revision or the next sub-strand. Every learner's notebook contains the agreed Final Answer sentence, which serves as a portable exit reference. |
| <b>Model Building Phase</b><br> <b>VIDEO:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Model Revision (10 min): Learners open their notebooks to their Lesson 1 naïve model — the drawing they made at the very                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | For the Model Revision, say: 'Open to your Lesson 1 model. Look at it carefully. In RED pen, add everything you now know that                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Then-vs-Now Model Comparison with Peer Gallery Walk: comparing the Lesson 1 naïve model to the fully annotated final model makes                                                                                                                                                                                                                                                                             | Observable indicator: each learner's revised model contains all three formulae with labelled diagrams and a complete 'Then vs.                                                                                                                                                                                                             |

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| <p><a href="#">Impact of increasing the radius</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b></p> <p><a href="#">Area of Sectors and Segments (Flexbook)</a></p> <p>Source: CK-12</p> <p><a href="#">Search ARES for similar readings</a></p> | <p>start of the unit showing 'what they thought determined the size of a watermelon slice'. Each learner individually annotates their original Lesson 1 model in RED pen, adding: (a) the sector area formula with labels; (b) the triangle area formula with labels; (c) the segment area formula with labels; (d) an arrow from the 'angle' in the diagram labelled 'THIS is what changes the area — not just how it looks'. Below the revised model, learners write a 'Then vs. Now' statement: 'At the start of this unit I thought... Now I know...'. Gallery Walk (10 min): Each group posts their manila paper (from Phase 3) and their Group Data Table on the classroom wall. Groups rotate around the room (2 minutes per station) leaving one sticky-note comment per station using the sentence starter: 'One thing this group's work helped me understand about the watermelon puzzle is...'. The lesson closes with the teacher reading 3–4 sticky-note comments aloud and asking the whole class: 'Has Mama Wanjiku at Wakulima Market been treating her customers fairly? What would you tell her to do differently, using mathematics?'</p> | <p>your Lesson 1 self did not know. You have 10 minutes — no talking, this is individual work.' Circulate and look specifically for learners who only add formulas without annotating the diagram — prompt these learners: 'Can you point to WHERE in your drawing the angle <math>\theta</math> lives? Where is <math>r</math>? Where would the chord go to create a segment?' For the Gallery Walk, set a visible timer for 2-minute rotations. Insist on the sentence starter — do not accept 'good job' style comments. After the Gallery Walk, read 3–4 sticky notes aloud and ask: 'What would you tell Mama Wanjiku?' Wait 10 seconds, then cold-call 3 learners. Aim for a response that includes a specific recommendation (e.g., 'She should cut all wedges at the same central angle, or advertise the area in <math>\text{cm}^2</math> on a price board so customers can compare'). Close by saying: 'You have just done what mathematicians and market inspectors do — you used exact measurement to replace guessing with fairness. That is the power of area of part of a circle.'</p> | <p>conceptual growth concrete and visible — a key metacognitive practice in CBE. The Gallery Walk distributes the explanatory load across all groups, allowing learners to encounter four different worked examples and articulate what each contributed to their understanding of the driving question. The final 'advice to Mama Wanjiku' question requires learners to translate mathematical knowledge into a practical, Kenya-grounded recommendation — completing the full sense-making cycle from phenomenon to formula to application.</p> | <p>Now' statement of at least two sentences. Teacher collects the Group Data Tables as a summative artefact for the Observation Schedule assessment method (KICD Appendix 1). The 'advice to Mama Wanjiku' responses are used as an oral exit check — learners who cannot connect the formula to a practical recommendation are identified for follow-up in the next lesson's opening review.</p> |
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## D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your journal:

1. FORMULA DERIVATION: Did all groups successfully derive Area of segment =  $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$  independently, or did some groups require the subtraction idea to be prompted? If prompting was needed, which step (isolating the triangle vs. applying  $\frac{1}{2}r^2\sin\theta$ ) caused the most difficulty? How will this inform your instruction in Sub-Strand 2.6 (Surface Area of Solids) where similar composite-shape strategies apply?
2. CUSTOMER COMPLAINT CARD: Were learners able to correctly interpret why Piece D ( $\theta = 120^\circ$ ) has a larger segment area than Piece C ( $\theta = 90^\circ$ ), and articulate the role of  $\sin\theta$ ? If learners simply said 'bigger angle = bigger segment' without unpacking the  $\sin\theta$  term, note this as a gap in algebraic reasoning to address during revision.

3. DQB AUDIT: How many cards remained RED or YELLOW at the end? Were these extension questions posted by high-attaining learners, or foundational questions from learners who still struggled with the core formulae? Use this evidence to plan differentiated revision activities.
4. MODEL COMPARISON: Did learners' 'Then vs. Now' statements show genuine conceptual growth, or did they simply copy formulas without articulating a change in understanding? Select 3–4 representative notebooks (one from each attainment band) and use these as evidence in your portfolio for peer-learning discussions.
5. EQUITY AND FAIRNESS: The watermelon phenomenon was designed to connect mathematics to fairness and social justice — core values in the KICD framework. Did learners make this connection explicitly in their 'advice to Mama Wanjiku' responses? If not, how might you embed values-based connections more explicitly in future capstone lessons?
6. TIMING: The lesson is planned for 80 minutes. Note which phase ran over or under time and adjust the Observation Schedule assessment to be completed outside class if the Gallery Walk was rushed.

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>What did I observe?</b>                   | Learners physically cut the triangle (formed by two radii and the chord) from their sector cut-outs to isolate the curved segment, measured dimensions, and completed a Group Data Table recording sector area, triangle area ( $\frac{1}{2}r^2\sin\theta$ ), and segment area for four different central angles ( $40^\circ$ , $65^\circ$ , $90^\circ$ , $120^\circ$ ) using $r = 18$ cm — modelling a new Wakulima Market watermelon.                                                                                                                                                                                             |
| <b>What did I learn?</b>                     | Learners derived and applied the formula Area of segment = $(\theta/360)\pi r^2 - \frac{1}{2}r^2\sin\theta$ , verified it numerically against their cut-and-subtract results, used it to adjudicate the Customer Complaint Card (calculating which of Mama Wanjiku's four customers received the largest curved segment), and connected both sector and segment formulae to real-life fair-sharing and design contexts in Kenya.                                                                                                                                                                                                    |
| <b>How does this explain the phenomenon?</b> | Learners produced a complete, evidence-based Final Explanation of the driving question — articulating that the area of any wedge (sector) or curved-sliver (segment) of a circle is determined precisely by the central angle $\theta$ and radius $r$ through their respective formulae, not by visual appearance alone. They revised their cumulative circle-area model (annotating all three formulae onto their Lesson 1 naïve drawing), completed the Driving Question Board audit, and gave Mama Wanjiku a mathematically justified recommendation for ensuring every customer at Wakulima Market receives a truly fair share. |

| DIFFERENTIATION AND INCLUSION       |                                                                                       |                                                                                  |
|-------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Learner Need                        | Support Strategies                                                                    | Extension Strategies                                                             |
| English Language Learners           | Provide bilingual glossaries; allow use of first language for initial model drawing.  | Write extended explanations of observations; lead group discussions in English.  |
| Students with learning difficulties | Provide structured note frames; pair with a supportive partner during investigations. | Mentor peers; create additional visual models to explain concepts.               |
| Gifted and advanced learners        | Access to additional reading materials; challenge questions during DQB.               | Lead model-building discussions; research and present extension topics to class. |